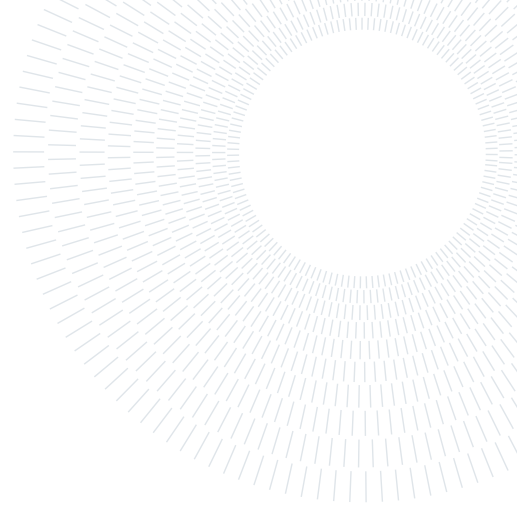




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SCUOLA DI INGEGNERIA INDUSTRIALE
E DELL'INFORMAZIONE



DATA SCIENCE

Final project

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Scope

This project study is a streamlined methanol production process (see figure) with strong simplification in the thermodynamic as well as the layout of the process. A dataset has been provided containing information derived with rigorous simulation of this process based on mass and energy conservation, van't Hoff chemical equilibrium for the reactor, Antoine vapor pressures and a Rachford–Rice flash for the separator (the four species involved are CO, H₂, CH₃OH and H₂O). Starting from that our goal is to create a surrogate model, to be used instead of the rigorous simulator for the downstream task whenever this would become the bottleneck part of the process in a real life plant.

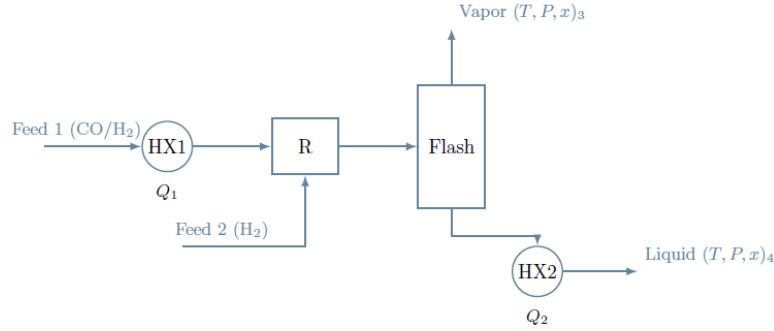


Figure 1: Simplified flow scheme

1. Exploratory data analysis

This first section is focused on the inspection of the data, through the use of different techniques for data visualization we're looking to identify correlations.

Starting from the scatter plots for inputs/outputs to the underlying potential correlation between the different variables. Specifically for each one of the outputs the connection with every input is studied.

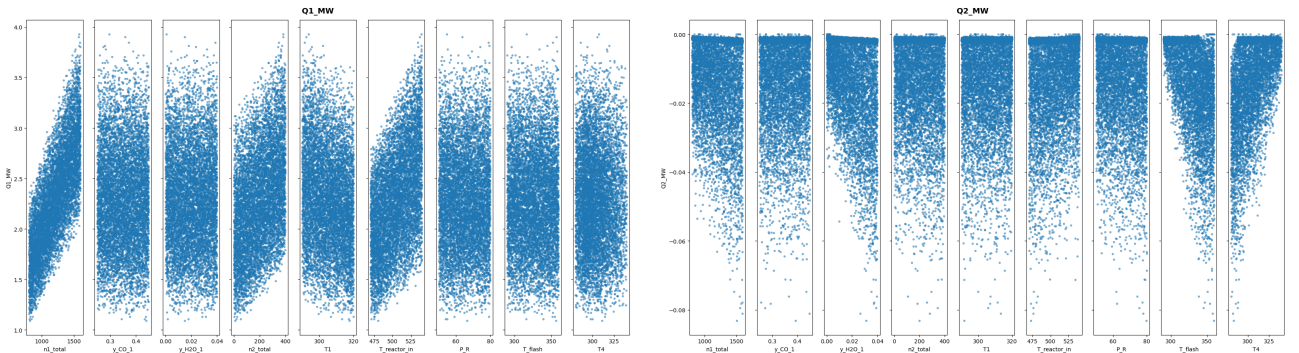


Figure 2: Heat duties of HX1 and HX2.

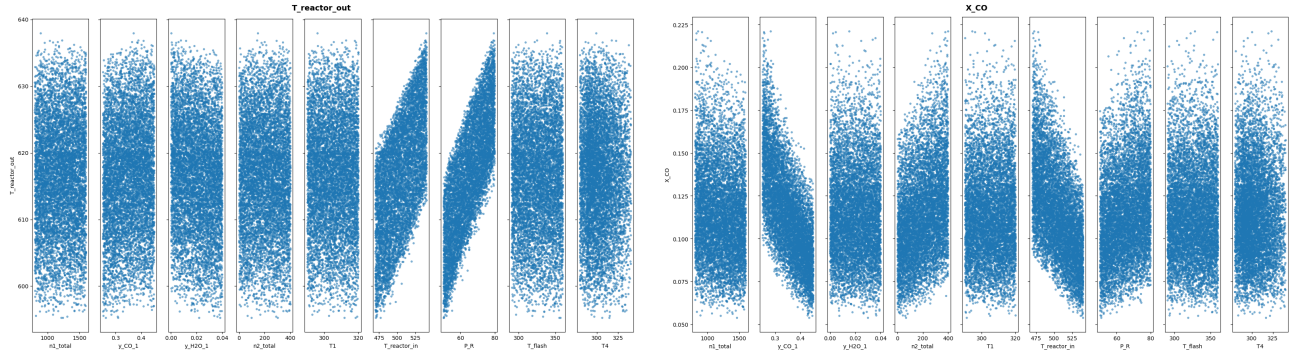


Figure 3: Adiabatic reactor outlet temperature and CO conversion.

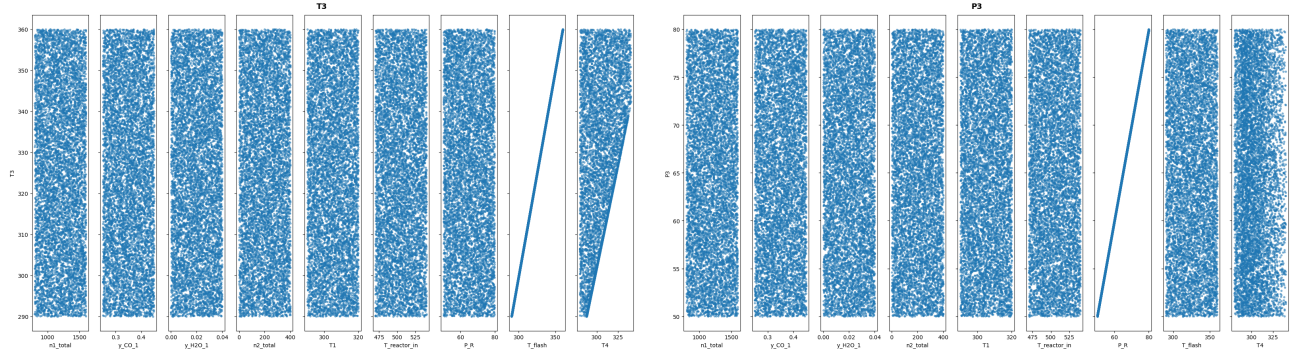


Figure 4: Temperature and pressure of stream 3.

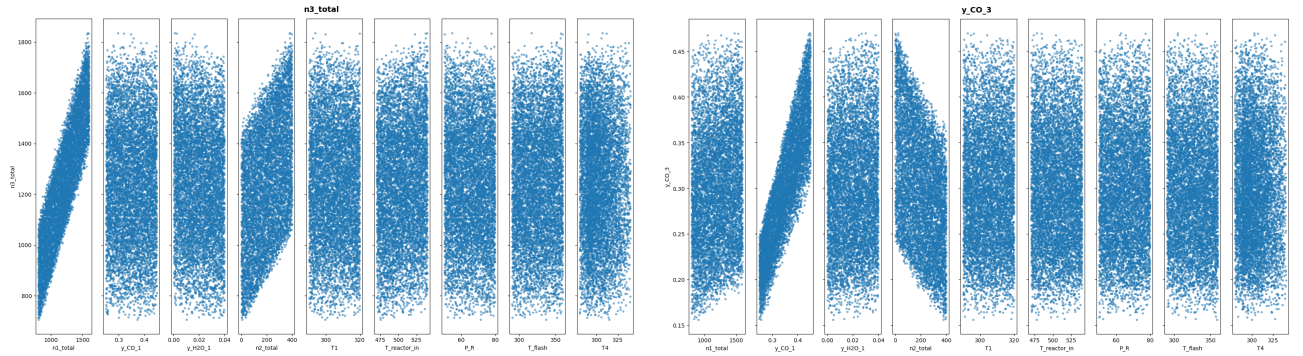


Figure 5: Total molar flow and CO mole fraction in stream 3.

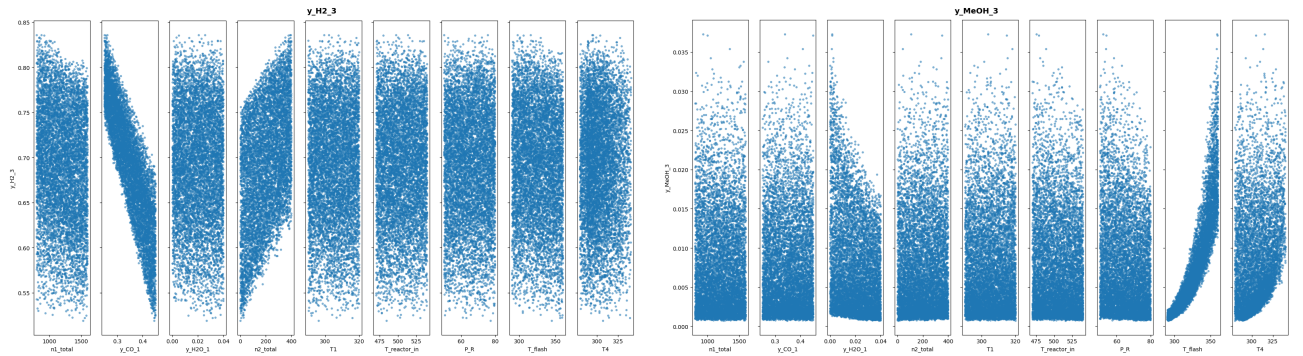


Figure 6: H₂ and methanol mole fractions in stream 3.

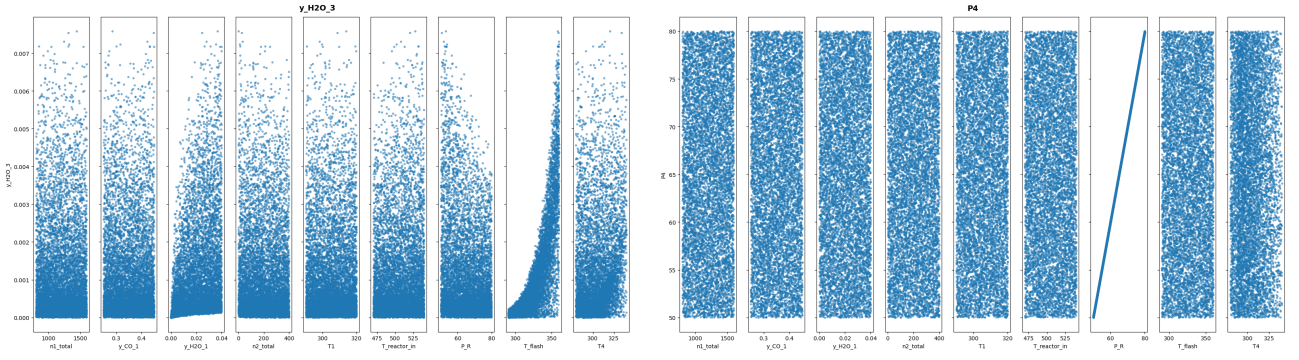


Figure 7: H₂O mole fraction in stream 3 and pressure of stream 4.

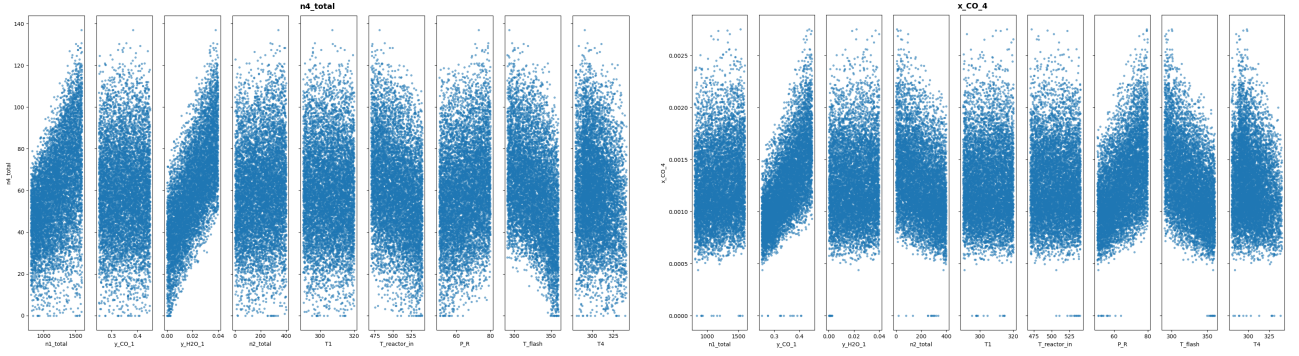


Figure 8: Total molar flow and CO mole fraction in stream 4.

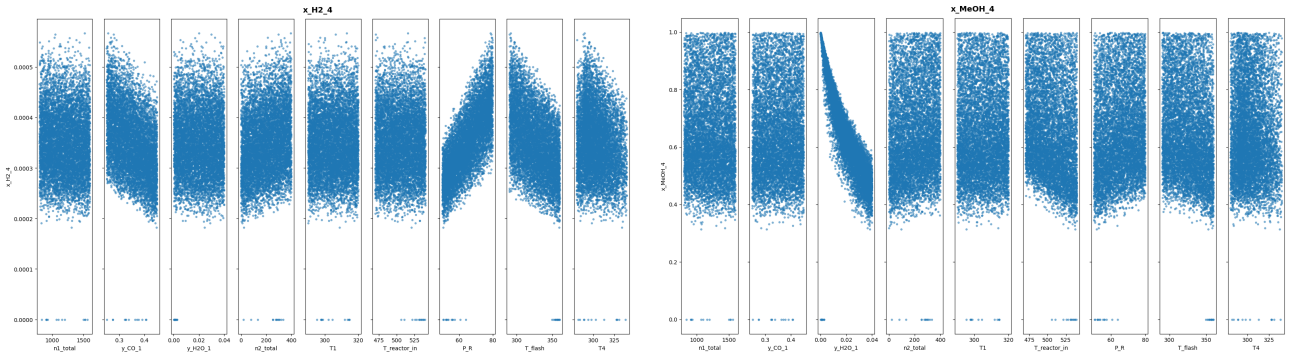


Figure 9: H₂ and methanol mole fractions in stream 4.

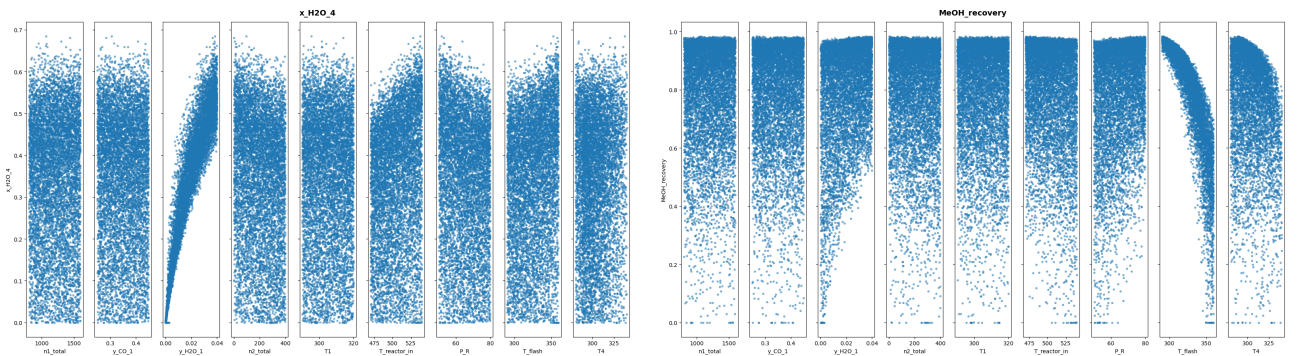


Figure 10: H₂O mole fraction in stream 4 and methanol recovery.

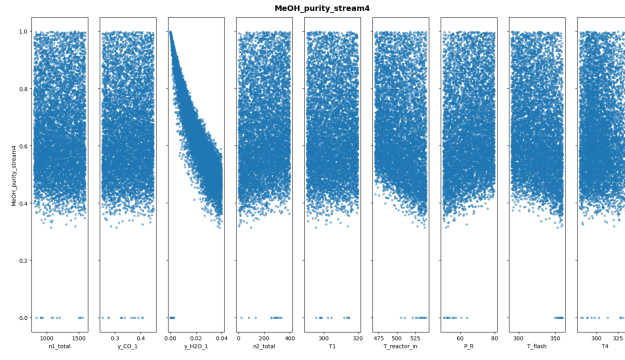


Figure 11: Methanol mole fraction in stream 4.

Starting from the duty of the first heater Q_1 we can see that there is an obvious almost linear positive dependency from the inlet stream n_1 and negative linear dependency for the temperature of that stream T_1 . The same logic for the temperature required at the inlet of the reactor and the number of moles of the second stream, increasing one of them will require and increase in the power supplied. For the second heat exchanger, the cooler after the flash, we see a positive dependency on the molar fraction of water entering in the fresh feed 1, and the temperature of the flash, an increase in one of the two values lead to an increase in the power requested. Opposite trend is shown for the temperature of the liquid product stream. For the outlet temperature of reactor the two marked correlations are with the inlet temperature and on the pressure required for the flash. The conversion of the CO in the reactor is clearly correlated to the inlet molar fraction, the molar flow and the temperature of the reactor following the thermodynamic laws. The temperature and pressure of the vapor flow exiting the flash are the same as the one imposed in the flash and therefore are not of interest to study, while the molar flow of the vapor is increasing with the increase of the two inlet streams. For the molar fraction of CO and H_2 in stream 3, these variables represent the un-reacted reagents, here we can see the effect of the molar flow of stream 2 (inversely related for the former and directly for the latter) and the molar fraction on stream 1 with opposite behavior. For the methanol fraction there is a strong dependency on the temperature of the flash, almost exponentially increasing. Looking now at the liquid stream exiting the flash we can see a increase related to the increase of molar fraction of water entering the process, since the water is not going in the gas phase, and for the same reason also the molar fraction of water in that stream is directly related to the molar fraction of water in stream 1. For the CO and H_2 we can see how the values are really compacted and in extremely small percentages, due to the high volatility of those 2 components that are present just as residues in this stream. The methanol, our product of interest, is effected mostly only by the molar fraction of water in stream 1, with a negative effect of course since it is diluting the product stream. Looking now at the 2 KPI methanol recovery and purity, the former is negatively affected by the flash temperature imposed because it drives the physical process of condensation by manipulating the vapor pressure of the chemical components. The water concentration in stream 1 also have an effect as "bottom line", it's not an extremely marked correlation but we can see from the graph that having higher $y_{H_2O_1}$ ensure to avoid extremely low values for methanol recovery. The purity is molar fraction in the liquid stream and so it's clearly correlated to the molar fraction of water in stream 1 since it's not a reactant.

Marginal Distributions of Inputs

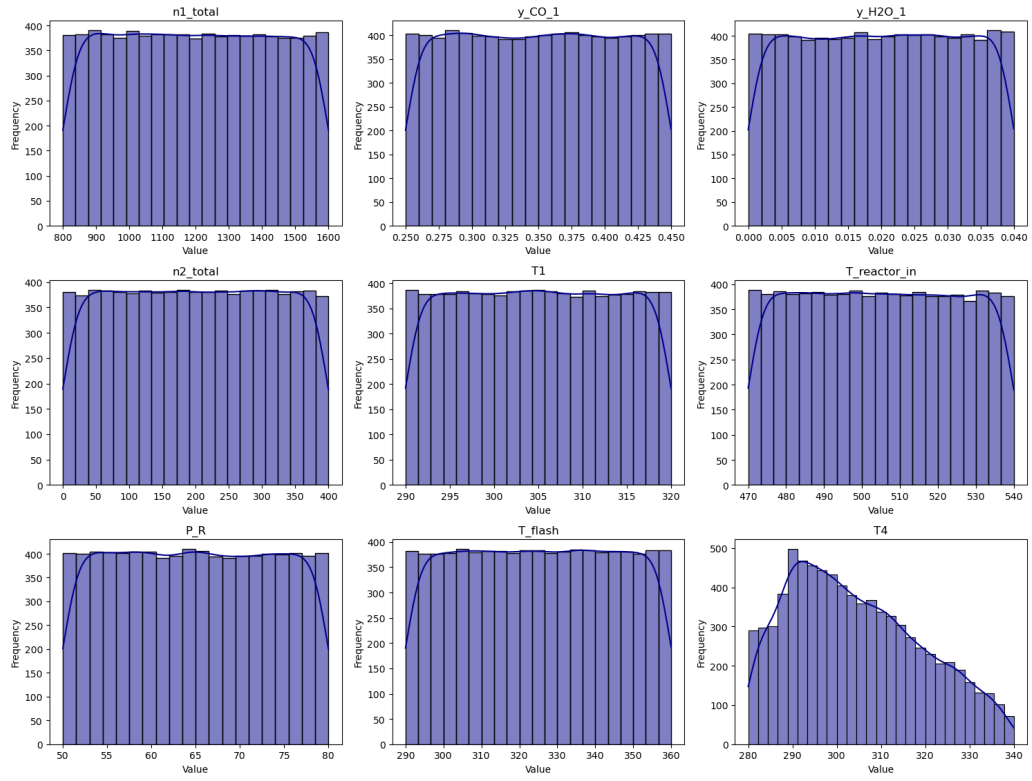


Figure 12: Marginal distribution of the inputs

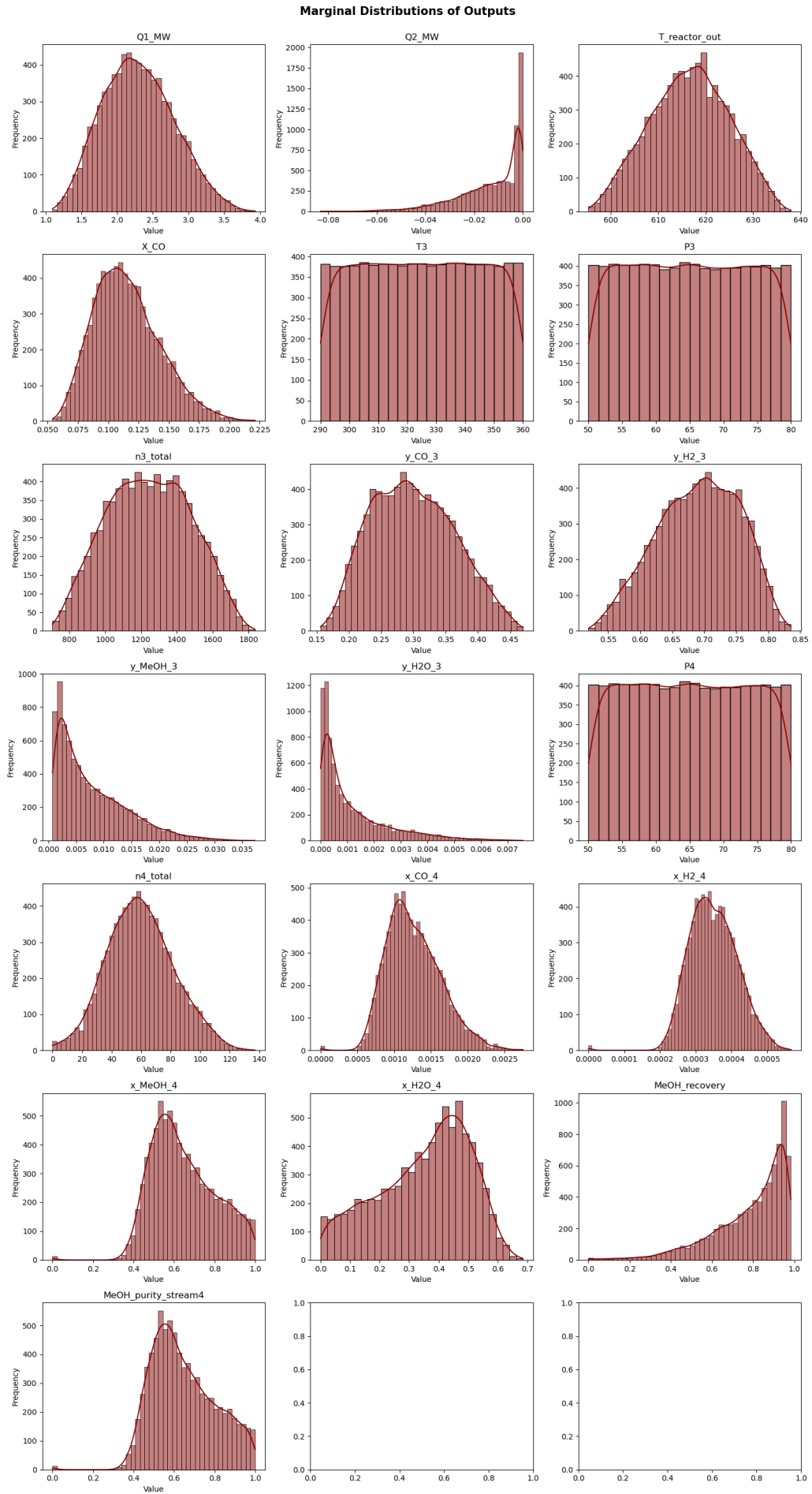


Figure 13: Marginal distribution of the outputs

The marginal distributions (histograms) for the four key process inputs display perfectly uniform, flat profiles across their defined operational bounds. This strict uniformity confirms that the training dataset was successfully generated using Latin Hypercube Sampling (LHS). Unlike the inputs, the marginal distributions of the process outputs are highly skewed, non-uniform, and exhibit extreme spikes at specific bounds. The most critical feature is the massive zero-inflation observed in all variables associated with the liquid product stream. Correspondingly, variables associated with the vapor stream (like $n_{3,tot}$) exhibit maximum-value spikes, representing conditions where the totality of the mass flow exits as vapor. This is a degenerate output, these zero-inflated distributions perfectly quantify the frequency of a single-phase vapor flash. When the combination of uniformly sampled P_R and T_{flash} results in a state that remains above the mixture’s dew point, no condensation occurs. All the synthesized methanol, water, and un-reacted syngas remain in the vapor phase (stream 3). Consequently, stream 4 ceases to exist, plunging its associated flow rate, recovery, and purity metrics to absolute zero. The extreme skewness and presence of hard physical boundaries (cliffs at zero) prove that simple multivariate linear regression will be fundamentally inadequate for predicting liquid-phase outputs.

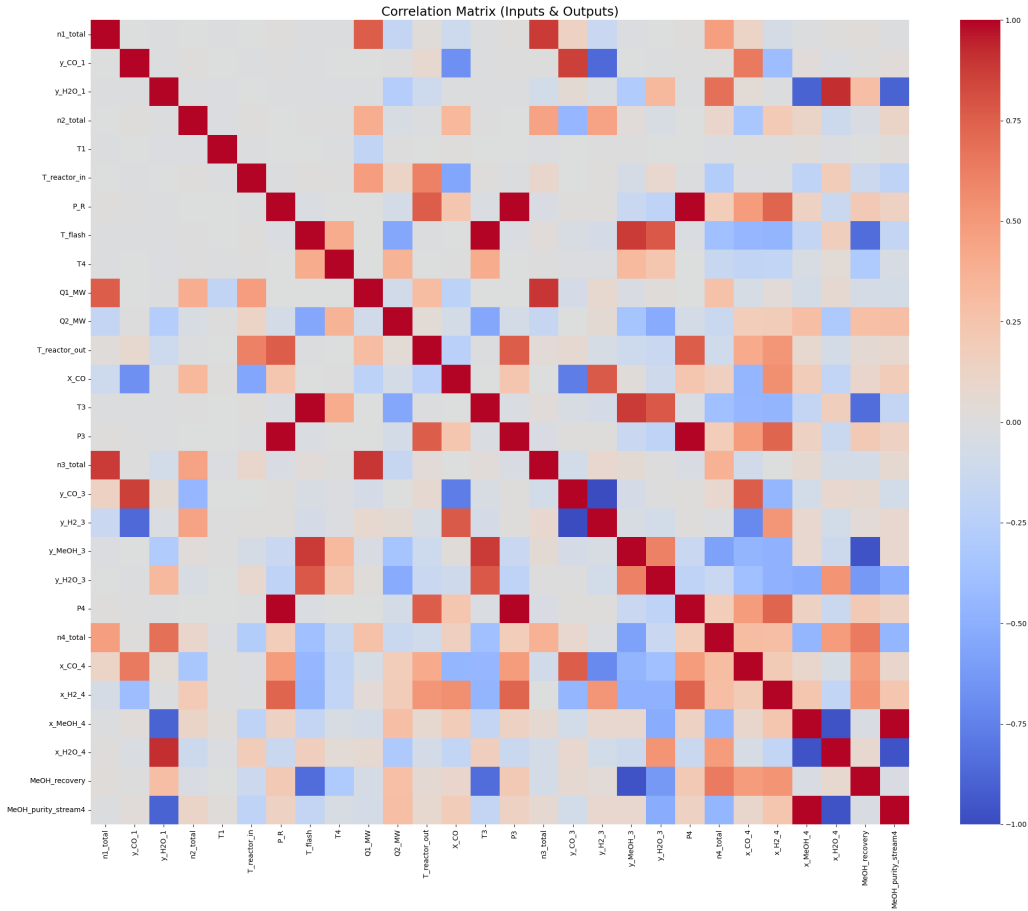


Figure 14: Correlation matrix

By scanning this heatmap, we can immediately divide the behavior of the chemical plant into two categories: simple deterministic rules (which show up as strong correlations) and complex thermodynamic phenomena (which show up as weak or zero linear correlations despite known physical causality). The former are the values connected by the physicality of the model (example the flash temperature

and temperature of stream 3). The latter are the ones we’re interested in, a first observation can be that primary process drivers like the pre-heater duty (Q_1) and flash temperature (T_{flash}) show very low Pearson correlation coefficients with critical performance metrics like methanol recovery, $n_{4,tot}$, or X_{CO} . This is coherent with the non linearity of the relationship between the heating via Q_1 and the production of methanol since there is an optimal temperature range. A second observation can be that the post-flash cooler duty (Q_2) shows practically zero correlation with reactor outputs, stream 3 properties, or the total mass flow of stream 4.

We calculated the variance of all the variables to check if some of them are almost constant to be able to remove them from the neural network training since those would not provide any useful information to the model. The four identified are:

Table 1: Variables identified as almost constant.

Variable	Variance	Standard deviation
$y_{MeOH,3}$	3.89×10^{-5}	6.24×10^{-3}
$y_{H_2O,3}$	1.72×10^{-6}	1.31×10^{-3}
$x_{CO,4}$	1.34×10^{-7}	3.66×10^{-4}
$x_{H_2,4}$	4.37×10^{-9}	6.61×10^{-5}

2. Baseline model

The second task focused on the development of baseline multivariate regression models for the prediction of the process outputs from the available operating variables. The purpose of this step was to establish reference levels of predictive performance before introducing neural-network-based surrogate models.

Since the process variables are interconnected and several outputs depend simultaneously on the same set of inputs, the models were formulated as multivariate regression problems. Both a linear and a non-linear baseline approach were considered in order to assess the extent to which nonlinear relationships and interactions between process variables affect the prediction accuracy. The resulting baseline models provide a benchmark against which the performance of the subsequent neural-network architectures can be evaluated.

2.1. Lasso Regression

Lasso regression was selected as the linear baseline model for the surrogate modeling task. It provides a simple and interpretable reference model while introducing L_1 regularization, which penalizes large regression coefficients and can force less relevant coefficients to zero. This property is particularly useful when several process inputs may have a limited contribution to a specific output variable.

For each process output, a separate Lasso regression model was fitted using the same set of nine input variables. Therefore, the overall surrogate produced predictions for all 19 outputs, while the regularization strength was independently selected for each output. Prior to model fitting, all input variables were standardized to zero mean and unit variance in order to ensure that the penalty applied by Lasso regression was not affected by the different numerical scales of the process variables.

For a generic output y_j , the Lasso regression coefficients were obtained by minimizing the following objective function:

$$\hat{\beta} * j = \arg \min * \beta_j \left[\frac{1}{N} |\mathbf{y}_j - \mathbf{X}\beta_j|_2^2 + \alpha |\beta_j|_1 \right], \quad (1)$$

where $\mathbf{X}$ is the matrix of standardized input variables, $\mathbf{y}_j$ is the vector of observations for output j , β_j is the coefficient vector, and α is the regularization parameter. A small value of α produces a model close to ordinary linear regression, whereas a large value increases coefficient shrinkage and generates a sparser model.

A logarithmic grid of 100 α values between 10^{-4} and 10^2 was tested for each output. The optimal value was selected using the Bayesian Information Criterion (BIC). The Akaike Information Criterion (AIC) was also calculated as an additional diagnostic metric. Both criteria account for the residual error and the effective complexity of the model:

$$AIC = N \ln \left(\frac{RSS}{N} \right) + 2k, \quad (2)$$

$$BIC = N \ln \left(\frac{RSS}{N} \right) + k \ln(N), \quad (3)$$

where RSS is the residual sum of squares, N is the number of observations, and k is the number of active model parameters, calculated as the number of non-zero Lasso coefficients plus the intercept. Lower AIC and BIC values indicate a more favourable compromise between fitting accuracy and model complexity. BIC was adopted as the selection criterion because it applies a stronger penalty to unnecessarily complex models.

Figure 15 shows a representative AIC and BIC trend as a function of the regularization parameter. Both criteria reach their lowest values in the low- α region and increase as α becomes larger. This indicates that excessive coefficient shrinkage removes relevant information from the model and leads to underfitting. The selected value of α for each output corresponds to the minimum BIC value.

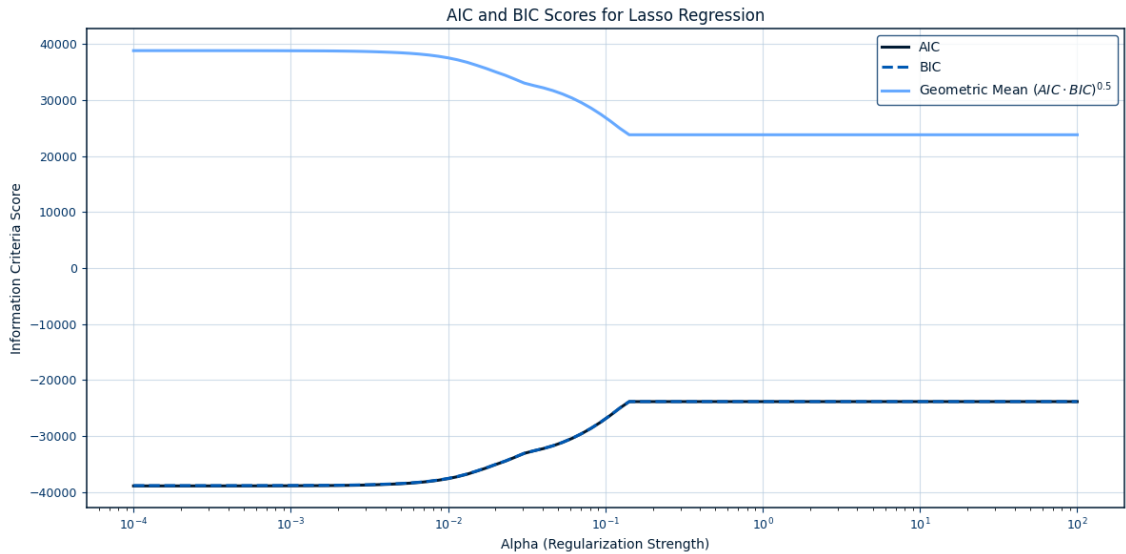


Figure 15: Representative AIC and BIC trends.

Table 2: Average performance metrics of the final Lasso regression model.

Dataset	Mean MAE	Mean RMSE	Mean R^2	Median R^2	Minimum R^2
Validation	0.481183	0.663508	0.86969	0.95090	-0.00197
Test	0.486102	0.656894	0.86907	0.95053	-0.00005

As shown in Table 2, the Lasso Regression model achieved mean R^2 values of 0.86969 and 0.86907 on the validation and test datasets, respectively. The corresponding median R^2 values are close to 0.95 for both datasets. The difference between the mean and median values indicates that the model predicts many outputs effectively, while a limited number of variables are not adequately represented by a sparse linear relationship.

The validation and test metrics are very similar. The mean MAE changes from 0.481183 on the validation dataset to 0.486102 on the test dataset, while the mean RMSE changes from 0.663508 to 0.656894. This close agreement indicates that the model has stable generalization behaviour and that the validation results are representative of its performance on previously unseen data.

Figure 16 reports the individual R^2 , RMSE, and MAE values for all process outputs. The largest absolute MAE and RMSE values are associated with $n_{3,\text{total}}$ and $n_{4,\text{total}}$, as well as, to a smaller extent, with the reactor outlet temperature. These variables have substantially wider numerical ranges than composition variables, pressures, and conversion-related outputs. Therefore, the absolute error values should be interpreted together with the corresponding R^2 values.

The Lasso model accurately captures several variables, including the heat duties, reactor outlet temperature, CO conversion, stream temperatures, pressures, total vapour flow rate, and the main vapour-phase composition variables. Their validation and test R^2 values remain close to unity, indicating that their variation can be approximated reasonably well by linear relationships within the investigated operating range.

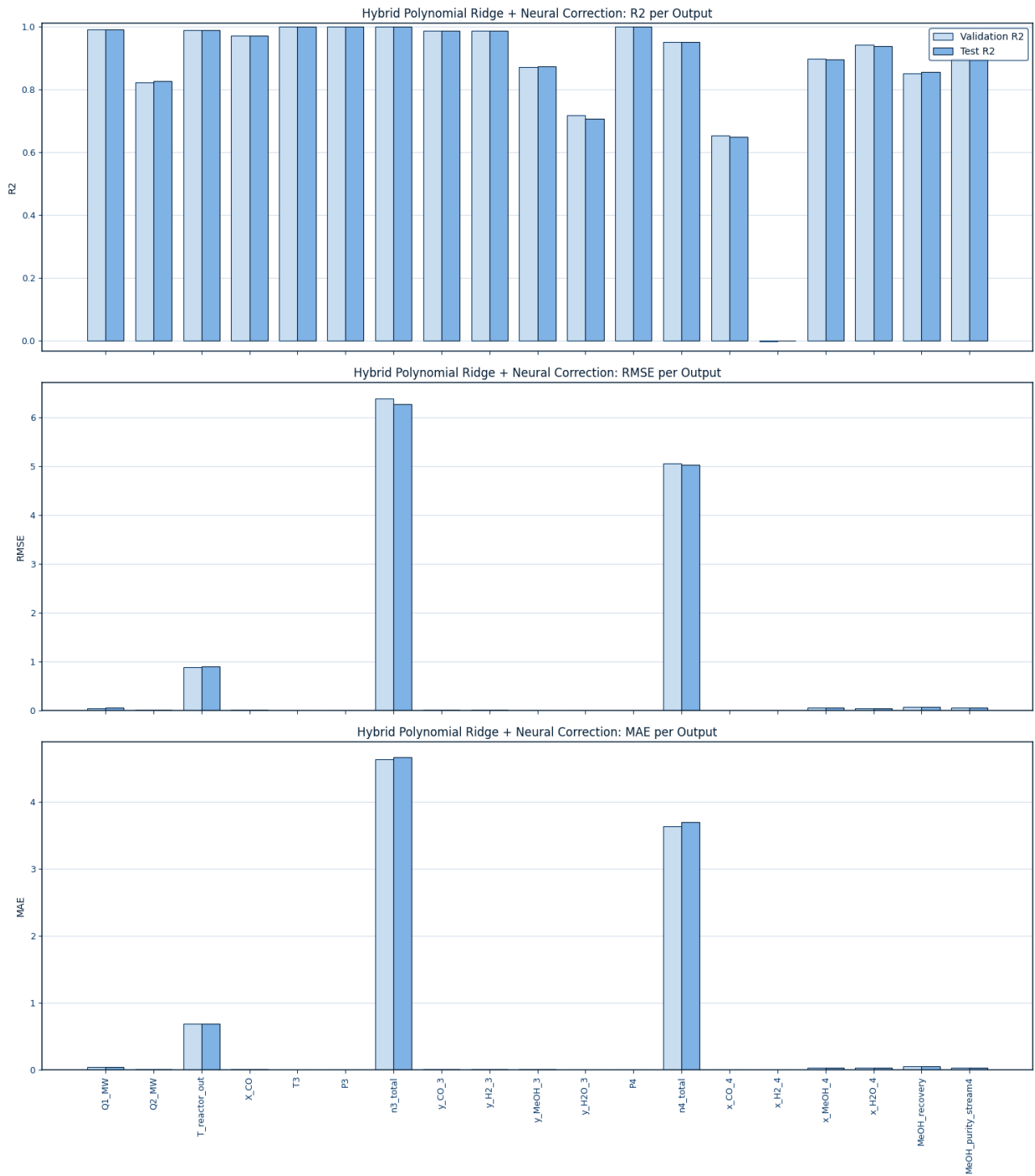


Figure 16: R^2 , RMSE, and MAE values for the Lasso Regression model per output.

The parity plots in Figure 17 illustrate the remaining limitations of the linear baseline. For the well-predicted variables, the points are closely distributed around the ideal diagonal line. Lower predictive accuracy is observed for variables influenced by the flash separation stage and phase equilibrium, including $x_{H_2,4}$, $x_{CO,4}$, $y_{H_2O,3}$, methanol recovery, and methanol purity. In particular, $x_{H_2,4}$ has an R^2 value close to zero, indicating that its variation is not captured adequately by the Lasso model. These deviations are consistent with the nonlinear and regime-dependent vapour–liquid equilibrium behaviour of the flash separator, which cannot be fully described by a sparse linear surrogate. Overall, Lasso regression provides a useful and computationally inexpensive baseline predicting the

majority of the process outputs with good accuracy and shows consistent validation and test performance. However, its reduced performance for liquid-phase and flash-related variables highlights the need for nonlinear surrogate models, which are considered in the following sections.

Lasso Regression Parity Plots for Test Outputs

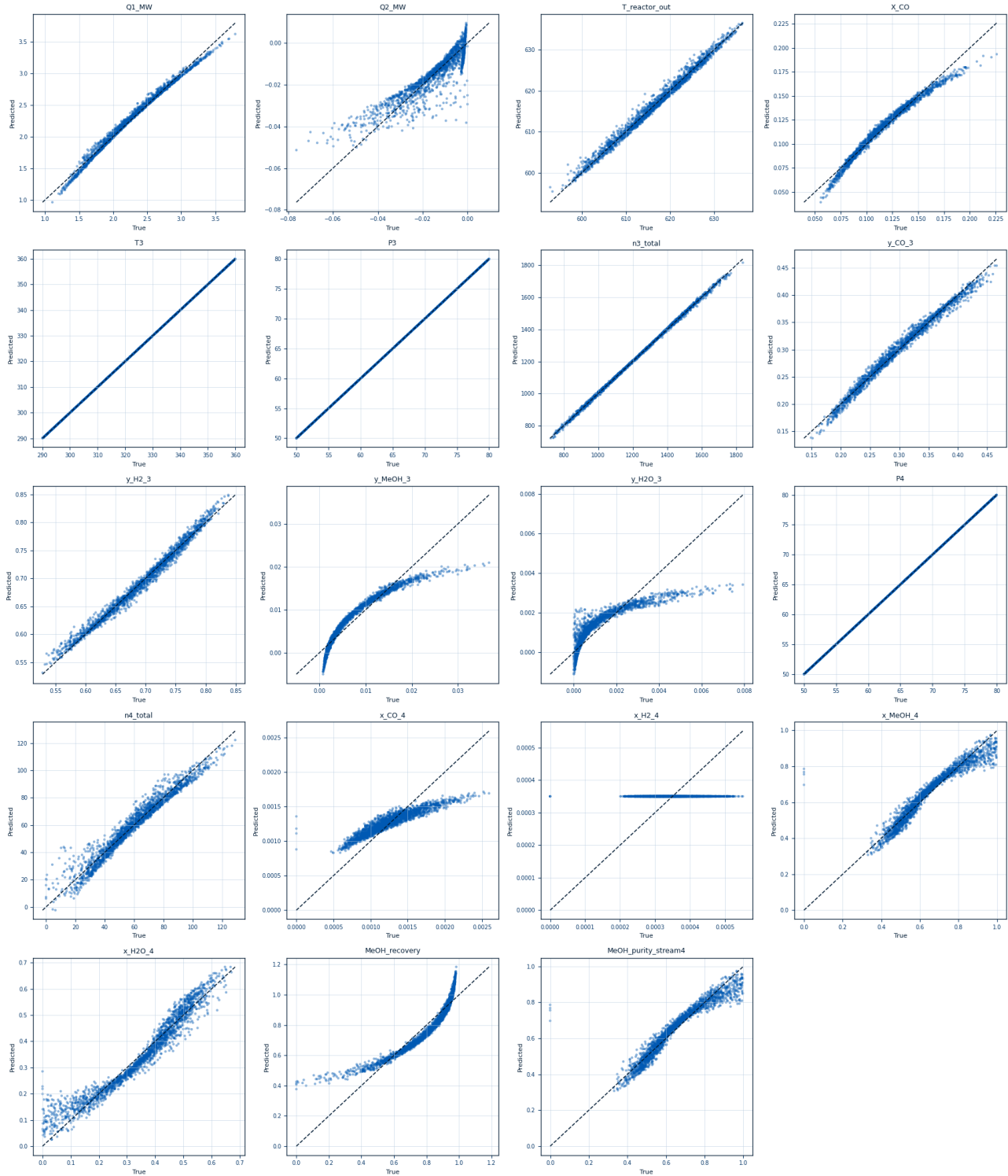


Figure 17: Parity plots for the Lasso regression model per output.

2.2. Polynomial Ridge Regression

Polynomial Ridge Regression was selected as the nonlinear baseline model for the surrogate modelling task: the method combines a polynomial expansion of the original input variables with Ridge regularization. In this way, the model remains linear with respect to the regression coefficients, but it is nonlinear with respect to the original process inputs. This makes it suitable for capturing smooth nonlinear relationships while preserving a relatively simple and interpretable mathematical structure. The general form of the model can be written as:

$$\hat{\mathbf{Y}} = \Phi(\mathbf{X})\mathbf{B}, \quad (4)$$

where $\mathbf{X}$ is the matrix of original input variables, $\Phi(\mathbf{X})$ is the polynomial feature matrix, $\mathbf{B}$ is the matrix of regression coefficients, and $\hat{\mathbf{Y}}$ is the matrix of predicted outputs.

The Ridge regression coefficients are obtained by minimizing the following objective function:

$$J(\mathbf{B}) = \sum_{i=1}^N \|\mathbf{y}_i - \hat{\mathbf{y}}_i\|_2^2 + \alpha \|\mathbf{B}\|_2^2, \quad (5)$$

where the first term represents the prediction error and the second term penalizes large regression coefficients. The parameter α controls the strength of the regularization.

This method is well suited to the present dataset because the observed input-output relationships appear to be smooth and structured rather than highly noisy. This behaviour can be seen from the exploratory analysis and from the strong performance of low- to moderate-complexity regression models. Moreover, the number of input variables is limited, while the number of available samples is relatively large, resulting in polynomial terms that can capture relevant nonlinear interactions between process variables without requiring an excessively complex model.

Before model fitting, the input variables were standardized using `StandardScaler`. The output variables were also standardized through `TransformedTargetRegressor`. This step is important because the outputs have different physical meanings and numerical scales, including heat duties, temperatures, pressures, flowrates, mole fractions, recovery, and purity.

2.2.1 Cross-validation and model selection

The polynomial degree and Ridge regularization parameter were selected using 5-fold cross-validation. In this procedure, the training set is divided into five subsets, and, for each hyperparameter combination, the model is trained on four subsets and validated on the remaining one. This process is repeated five times, so that each subset is used once as validation data. The final cross-validation score is then obtained by averaging the validation errors over the five folds.

The metric used for hyperparameter selection was the mean normalized RMSE across all output variables:

$$\text{Mean nRMSE} = \frac{1}{m} \sum_{j=1}^m \frac{\sqrt{\frac{1}{N} \sum_{i=1}^N (y_{i,j} - \hat{y}_{i,j})^2}}{y_{j,\max} - y_{j,\min}}, \quad (6)$$

where $m = 19$ is the number of output variables. The normalization makes the metric comparable

across outputs with different units and magnitudes.

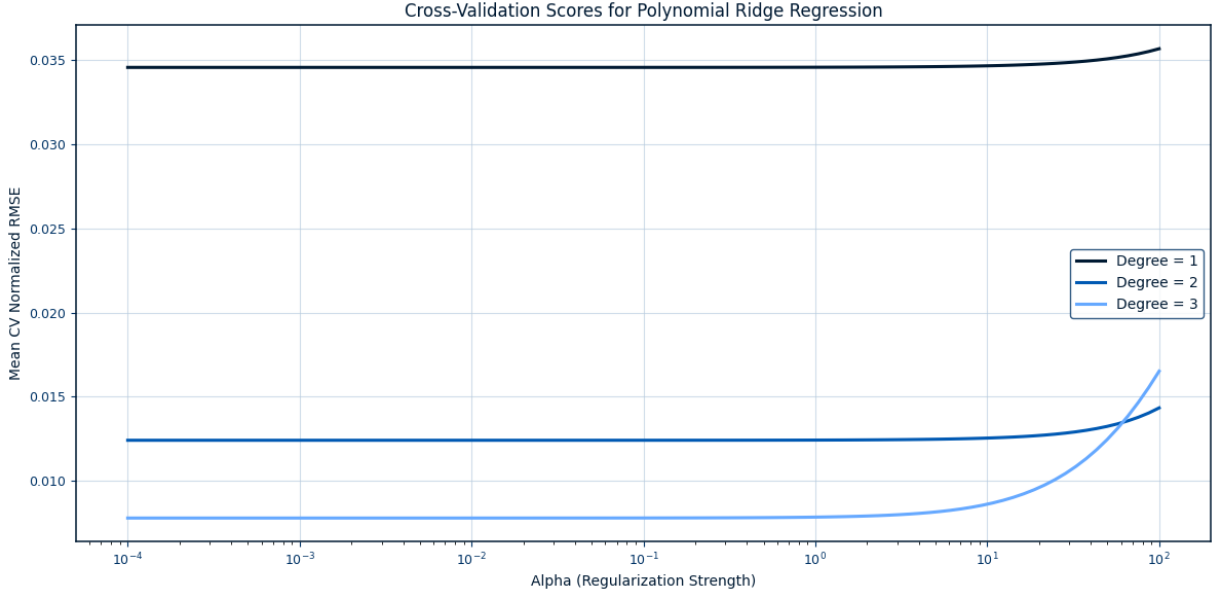


Figure 18: Cross-validation scores for Polynomial Ridge Regression.

Figure 18 shows that the linear model, corresponding to polynomial degree 1, gives the highest cross-validation error, indicating that a purely linear relationship is not sufficient to describe the process behaviour. Introducing second-order terms significantly reduces the error, showing that quadratic effects and pairwise interactions between input variables are important. The best performance is obtained with a third-degree polynomial. This suggests that cubic terms provide additional flexibility and improve the approximation of the nonlinear simulator response.

The lowest error is obtained at the smallest tested value of α , meaning that only weak Ridge regularization is required. This result is consistent with the observed characteristics of the dataset: the input-output relationships appear to be smooth and not strongly affected by random noise, therefore strong coefficient shrinkage is not necessary. At high values of α , the cross-validation error increases, indicating that excessive regularization reduces the flexibility of the polynomial model and causes underfitting.

Table 3: Polynomial feature count and best cross-validation score for each polynomial degree.

Degree	Number of polynomial features	Best CV normalized RMSE
1	9	≈ 0.0345
2	54	≈ 0.0124
3	219	≈ 0.0078

Based on the cross-validation results, the final Polynomial Ridge model was built using a third-degree polynomial expansion, that with 9 original input variables, this expansion generates 219 polynomial features. The resulting feature space includes first-order terms, second-order interaction and squared terms, and third-order cubic and mixed interaction terms.

In terms of the actual process variables, examples of possible polynomial features are:

$$T_{\text{flash}}, \quad P_R, \quad T_{\text{flash}}^2, \quad T_{\text{flash}}P_R, \quad P_R^2, \quad T_{\text{flash}}^3, \quad T_{\text{flash}}^2P_R, \quad T_{\text{flash}}P_RT_4. \quad (7)$$

Therefore, the final model can account not only for the direct effect of each input variable, but also for nonlinear coupling effects between operating conditions.

2.2.2 Prediction results

After hyperparameter selection, the final Polynomial Ridge model was retrained using the complete training dataset and evaluated on the independent test dataset. The model performance was assessed using the coefficient of determination (R^2), mean absolute error (MAE), and root mean squared error (RMSE).

Table 4: Average performance metrics of the final Polynomial Ridge model.

Dataset	Mean MAE	Mean RMSE	Mean R^2	Median R^2	Minimum R^2
Validation	0.022926	0.042965	0.99244	0.99976	0.95155
Test	0.023079	0.044670	0.99167	0.99974	0.94903

Table 4 shows that the Polynomial Ridge model achieved high predictive accuracy on both the validation and independent test datasets. The mean R^2 values are 0.99244 and 0.99167 for the validation and test sets, respectively, while the median R^2 values are close to unity. This indicates that the model accurately represents the behaviour of the majority of the process outputs.

The validation and test metrics are also closely aligned. The mean MAE changes only from 0.022926 on the validation dataset to 0.023079 on the test dataset, while the mean RMSE changes from 0.042965 to 0.044670. This agreement indicates that the selected model generalizes well to previously unseen process conditions and does not show substantial overfitting.

Figure 19 summarizes the individual R^2 , RMSE, and MAE values for all outputs. For most variables, the validation and test R^2 values are very close to one. In particular, the model accurately predicts the heat duties, reactor outlet temperature, CO conversion, stream temperatures, pressures, total vapour flow rate, and the main vapour-phase composition variables.

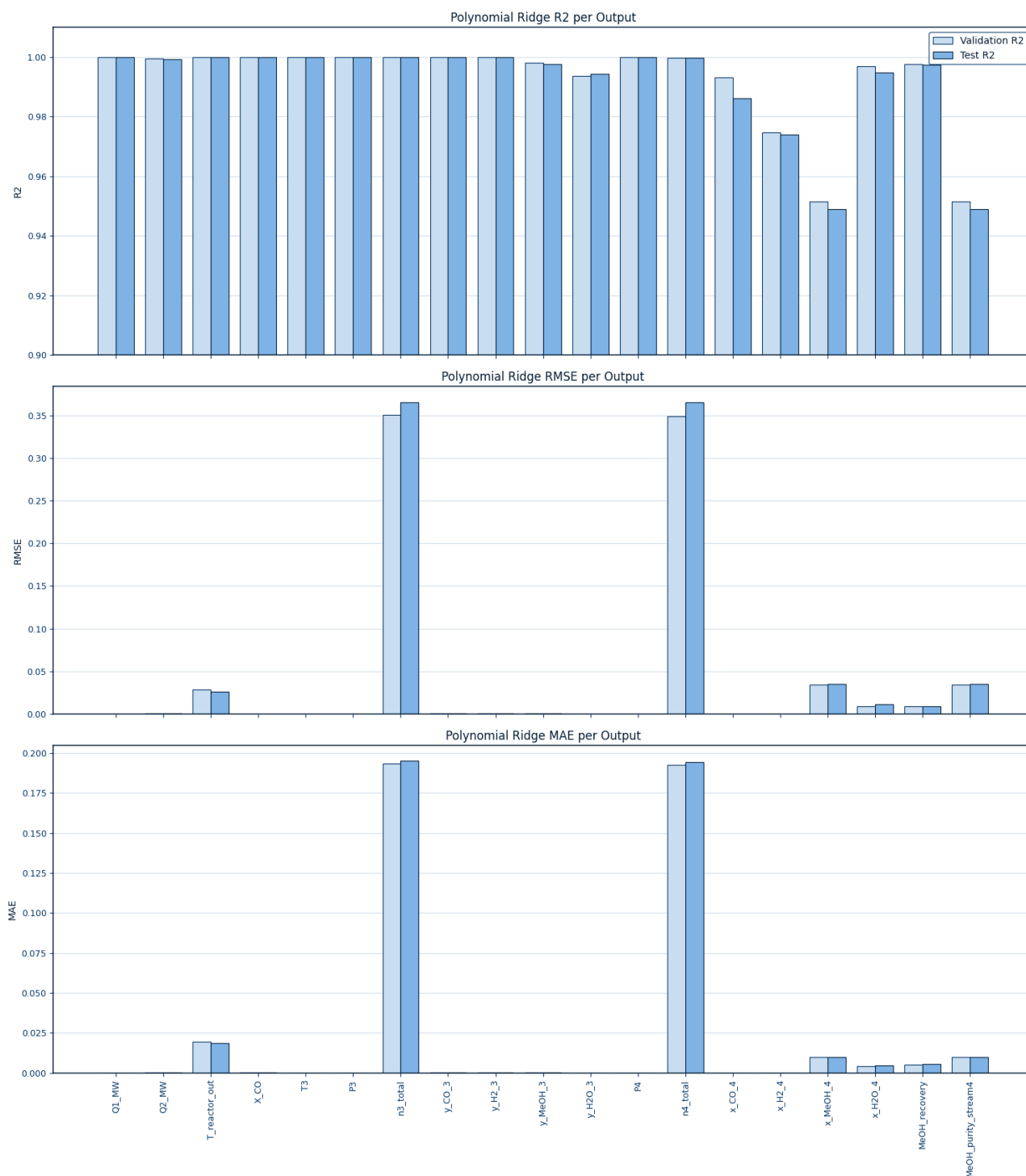


Figure 19: R^2 , RMSE, and MAE values for the Polynomial Ridge model per output.

The largest absolute RMSE and MAE values are associated with the total molar flow rates $n_{3,\text{total}}$ and $n_{4,\text{total}}$. This is expected because these variables have substantially larger numerical ranges than mole fractions, conversion, recovery, and purity variables. Therefore, the relatively higher absolute errors for the flow rates should be interpreted together with their corresponding R^2 values, which remain close to unity.

The lowest R^2 values are observed for selected liquid-stream variables, particularly $x_{\text{MeOH},4}$ and $\text{MeOH_purity_stream4}$, although their test R^2 values remain close to 0.95. Slightly lower performance is also observed for some low-concentration liquid-phase species. These variables are influenced

by the nonlinear vapour–liquid equilibrium behaviour of the flash separator, where relatively small changes in operating conditions can produce significant relative changes in liquid-phase composition. The parity plots in Figure 20 confirm the strong predictive performance of the Polynomial Ridge model on the final test dataset. For the main operating variables and upstream process outputs, including Q_1 , Q_2 , reactor outlet temperature, conversion, T_3 , P_3 , P_4 , and $n_{3,\text{total}}$, the predicted values lie almost exactly on the ideal diagonal line. Similarly, the model accurately reproduces the principal vapour-phase compositions, particularly $y_{\text{CO},3}$ and $y_{\text{H}_2,3}$.

The flash separator outputs are also described accurately over most of their operating ranges. In particular, $n_{4,\text{total}}$, $x_{\text{MeOH},4}$, $x_{\text{H}_2\text{O},4}$, methanol recovery, and product purity remain closely distributed around the diagonal line. A wider relative spread is visible for variables with very small absolute values, especially $x_{\text{CO},4}$, $x_{\text{H}_2,4}$, and $y_{\text{H}_2\text{O},3}$. In these cases, even a small absolute deviation can appear substantial because the output range is narrow.

A limited number of deviations remain for liquid-phase composition variables and methanol purity at the lower end of their respective ranges. This behaviour is consistent with the sharper nonlinearities introduced by vapour–liquid separation. Nevertheless, the overall concentration of points around the ideal diagonal line confirms that the third-degree Polynomial Ridge model captures both the main process trends and the dominant nonlinear interactions between the operating variables.

Polynomial Ridge Parity Plots for Outputs

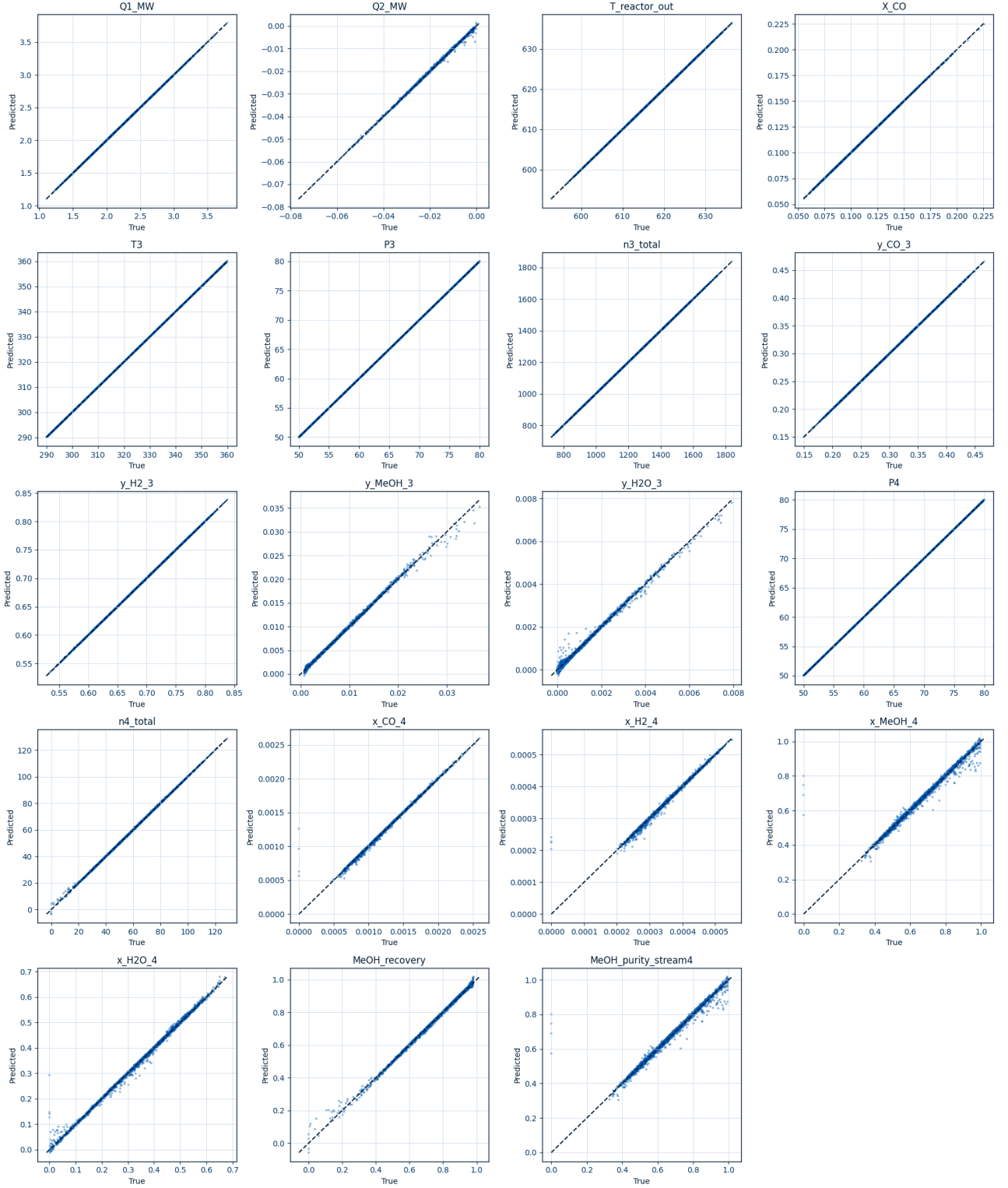


Figure 20: Parity plots for the Polynomial Ridge model per output.

3. Neural Network Surrogate Models

An alternative approach for process surrogation is the use of Artificial Neural Networks (NNs). Neural networks are data-driven models that can approximate complex relationships between process inputs and outputs without requiring an explicit analytical equation for the system. This makes them partic-

ularly suitable for chemical-process surrogate modelling, where the response may depend on multiple interacting operating variables and may exhibit strongly non-linear behaviour.

In this work, the neural networks are used as multivariate and multi-output regression models. The input layer receives the available process operating variables, while the output layer simultaneously predicts the complete set of process responses. During training, the model learns the relationship between inputs and outputs by adjusting its internal parameters to minimise a loss function based on the difference between predicted and reference values.

A feed-forward neural network is composed of successive layers. For a generic hidden layer, the input vector from the previous layer is first transformed through a weighted linear combination:

$$\mathbf{z}^{(l)} = \mathbf{W}^{(l)}\mathbf{a}^{(l-1)} + \mathbf{b}^{(l)}, \quad (8)$$

where $\mathbf{a}^{(l-1)}$ is the input from the preceding layer, $\mathbf{W}^{(l)}$ is the weight matrix, and $\mathbf{b}^{(l)}$ is the bias vector. The transformed signal is then passed through a non-linear activation function:

$$\mathbf{a}^{(l)} = f\left(\mathbf{z}^{(l)}\right), \quad (9)$$

where $f(\cdot)$ is the activation function. The activation function is essential because it allows the network to represent non-linear input-output relationships. Without non-linear activation functions, multiple layers would still be equivalent to a single linear transformation.

Each layer consists of several computational units, commonly referred to as neurons. Each neuron applies the weighted transformation and activation function to the information received from the previous layer. The number of hidden layers, the number of neurons in each layer, the activation functions, the learning rate, the batch size, and the regularization strategy are hyper-parameters that must be selected before training.

The weights and biases are not manually specified, they are learned from the training data through back-propagation and gradient-based optimization. In this procedure, the prediction error is propagated backwards through the network, allowing the optimizer to update each parameter in the direction that reduces the loss function.

The following sections describe the neural-network architectures considered in this work and evaluate their predictive performance relative to the baseline regression models.

3.1. NN Architectures

Three neural-network-based surrogate architectures were investigated in order to assess the effect of model capacity and hybrid modelling on the prediction of the process outputs. All architectures were formulated as multivariate, multi-output regression models: the nine available process inputs were used as network inputs, while the complete set of 19 process outputs was predicted simultaneously.

Before training, both input and output variables were standardized using the mean and standard deviation calculated from the training subset. This scaling procedure prevents variables with larger numerical ranges from dominating the training loss. The networks were trained by minimizing the mean squared error (MSE) between the predicted and reference scaled outputs. The Adam optimizer

was used for the direct neural-network models, while early stopping based on the validation loss was applied to retain the model weights corresponding to the best validation performance.

1. **Short Neural Network (SNN).** The first architecture was a relatively compact feed-forward neural network with one hidden layer composed of 64 neurons. The network structure can be summarized as:

$$9 \rightarrow 64 \rightarrow 19. \quad (10)$$

Each hidden layer contained 64 neurons and used the hyperbolic tangent activation function:

$$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}. \quad (11)$$

The output layer used a linear activation function in order to predict continuous process variables. This architecture was selected as a first non-linear neural-network model with a limited number of trainable parameters. Its purpose was to determine whether a relatively simple neural network could improve upon the baseline regression models without introducing excessive model complexity.

2. **Long Neural Network (LNN).** The second architecture was designed with a larger capacity in order to represent more complex non-linear relationships and interactions between the process variables. The network structure was:

$$9 \rightarrow 124 \rightarrow 124 \rightarrow 19. \quad (12)$$

The Long Neural Network contained two hidden layers, each composed of 124 neurons and followed by a hyperbolic tangent activation function. Compared with the Short Neural Network, this model has both greater depth and a larger number of neurons per layer. Consequently, it can represent a wider range of non-linear input-output relationships. However, the increased flexibility also increases the number of trainable parameters and therefore the risk of over-fitting. The comparison between the Short and Long architectures was used to evaluate whether the additional complexity produced a meaningful improvement in validation and test accuracy.

3. **Hybrid Polynomial Ridge-Neural Network Model.** The third architecture combined the previously developed Polynomial Ridge model with a neural-network residual correction. Rather than predicting the complete output vector directly, the Polynomial Ridge model first provided a baseline prediction:

$$\hat{\mathbf{Y}}_{\text{poly}} = f_{\text{PolyRidge}}(\mathbf{X}). \quad (13)$$

The neural network was then trained to predict the residual error between the true process outputs and the Polynomial Ridge prediction:

$$\mathbf{R} = \mathbf{Y} - \hat{\mathbf{Y}}_{\text{poly}}. \quad (14)$$

The final hybrid prediction was obtained by adding the neural-network residual correction to the Polynomial Ridge prediction:

$$\hat{\mathbf{Y}}_{\text{hybrid}} = \hat{\mathbf{Y}}_{\text{poly}} + \hat{\mathbf{R}}_{\text{NN}}. \quad (15)$$

The residual neural network consisted of one hidden layer with 64 neurons each:

$$9 \rightarrow 64 \rightarrow 19. \quad (16)$$

In this case, the SiLU activation function was employed in the hidden layers. The inputs and residual targets were standardized before training, and the residual network was optimized using AdamW with weight decay and early stopping.

The motivation for the hybrid architecture was to exploit the strengths of both approaches. Polynomial Ridge regression can capture the main smooth nonlinear trends and interactions between the operating variables, while the neural network can focus on the remaining prediction errors. Therefore, the neural network is not required to learn the complete process response from the beginning, but only the nonlinear residual behaviour not fully represented by the polynomial model. This approach may improve predictive performance while reducing the learning burden of the neural network.

3.2. NN important values

Other than the selection of the error, architecture and optimizing algorithm there are a series of different parameters that must be chose to train a Neural Network. These parameters have been decided using different criteria for each one and are essential for the reproducibility of the model.

Batch : To have a compromise between the required computing power for each epoch and the effective training progress a the DataLoader algorithm has been used, enabling shuffling and using a batch size of 32.

Epoch Patience: With the passing of epochs during training both validation and training losses diminished; however after an initial steep decrease the errors tend to reach a plateau and obscollate. Once the plateau is reached the error will not furtherly diminish but will have a stationary behaviour; to stop the training once the plateau is reached a patience number has been added, if, during the training, the loss function stops decreasing for a number of consecutive epochs higher than the patience number the program is sopped and the epoch with the less loss function is assumed as result of the training. In the present work the patience number has been fixed to 10 epochs.

All the important values are listed in Table 5.

Table 5: Important parameters for NN

Parameter	Value
Patience	10
Batch Size	32
Shuffle	True

3.3. NN Results

The neural-network models were assessed using the mean absolute error (MAE), root mean squared error (RMSE), and coefficient of determination (R^2). For the Short Neural Network and Long Neural Network, the training and validation subsets were used to monitor the learning process and select the best model weights. After the final model weights had been selected, parity plots were generated on the independent final test dataset in order to evaluate the predictive behaviour on previously unseen samples.

3.3.1 Short Neural Network

The Short Neural Network was trained using one hidden layers with 64 neurons per layer. Its learning behaviour is shown in Figure 21: both training and validation losses decrease rapidly during the first epochs and then converge towards a stable region. The close agreement between the two curves indicates that the model maintains stable performance during training.

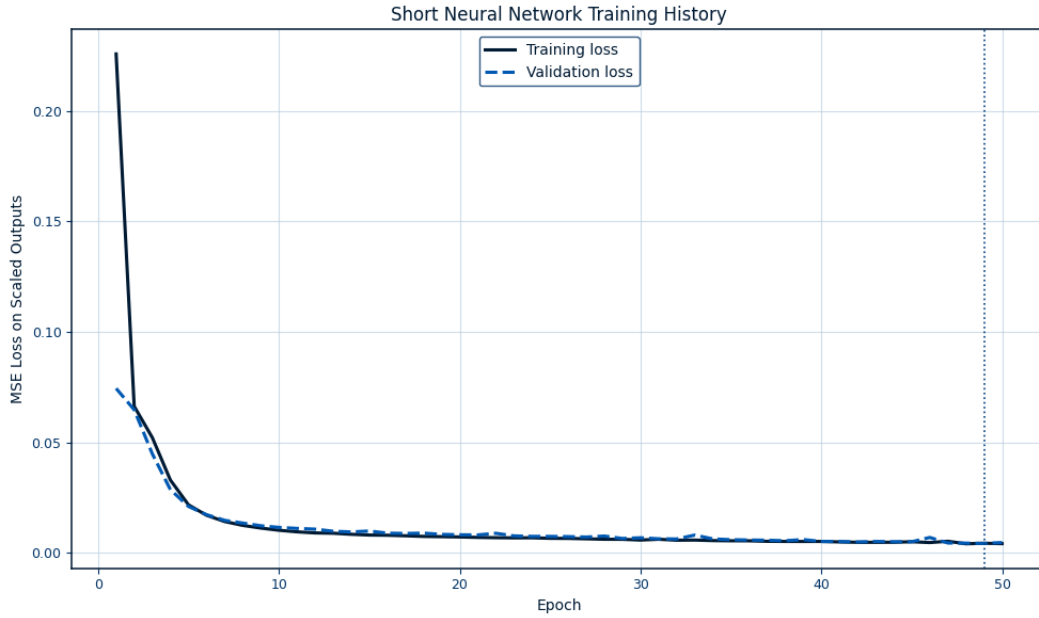


Figure 21: Training and validation loss histories for the Short Neural Network.

The average performance metrics are reported in Table 6. The model achieved a validation mean R^2 of 0.99604, while the median validation R^2 was 0.99902. The relatively high minimum R^2 value of 0.97877 indicates that the network provides consistent predictions across the complete set of outputs.

Table 6: Average performance metrics of the Short Neural Network.

Dataset	Mean MAE	Mean RMSE	Mean R^2	Median R^2	Minimum R^2
Training	0.3289451	0.379443	0.99604	0.99902	0.97877
Validation	0.2999623	0.0399264	0.99556	0.99899	0.97538

Figure 22 presents the R^2 , RMSE, and MAE values for each individual output. The network predicts the main process variables, including reactor temperature, pressure variables, vapour flow rate, and

the principal vapour-phase compositions, with high accuracy. Larger absolute errors are associated with $n_{3,\text{total}}$ and $n_{4,\text{total}}$, as these variables have substantially larger numerical ranges than the mole fractions and conversion-related outputs.

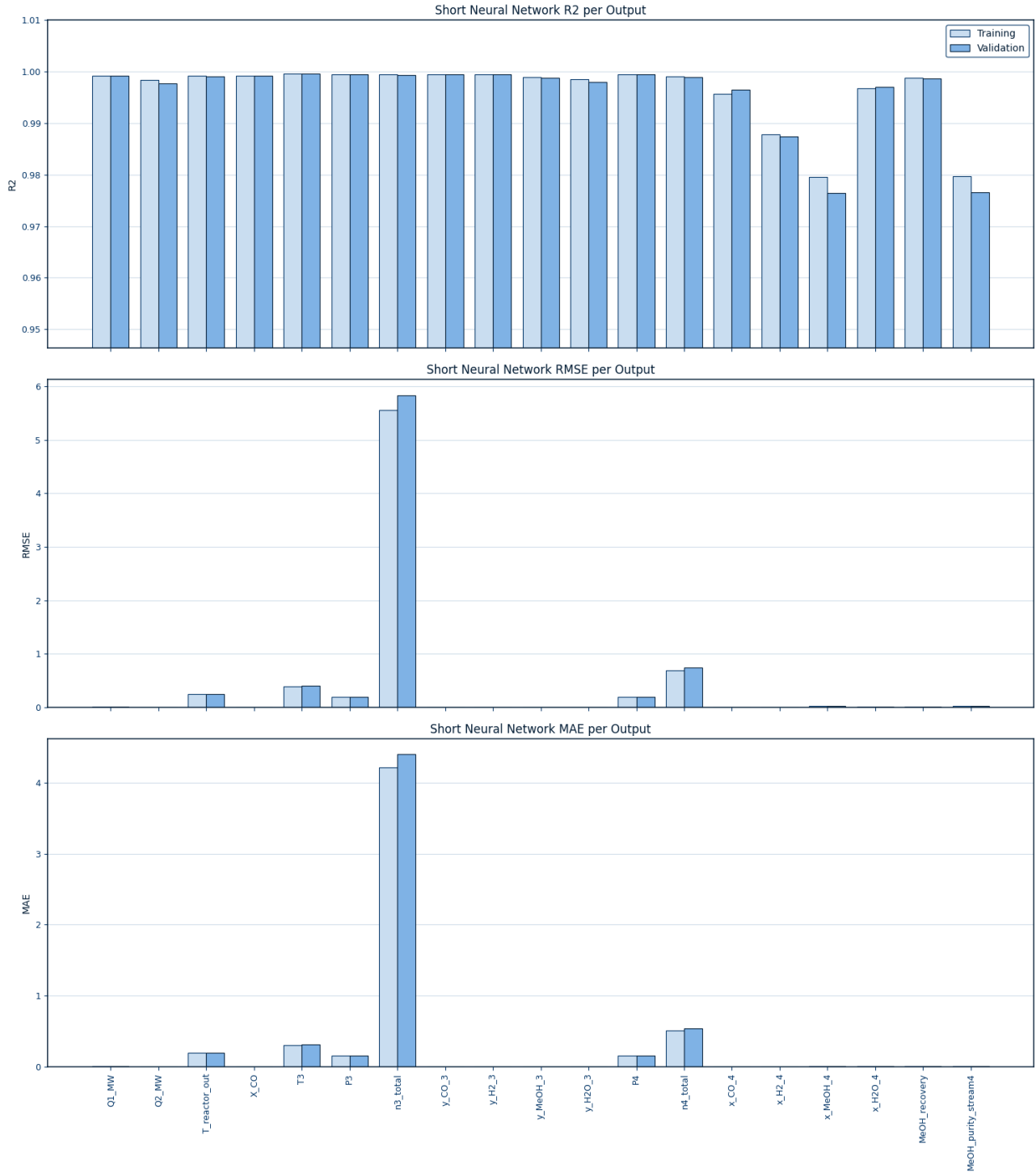


Figure 22: R^2 , RMSE, and MAE values for the Short Neural Network per output.

The validation parity plots are shown in Figure 23. Most points are closely distributed around the diagonal reference line, confirming accurate prediction of the main process trends. A larger dispersion is visible for very small liquid-phase mole fractions, particularly $x_{\text{CO},4}$ and $x_{\text{H}_2,4}$, as well as for a limited number of low-value methanol recovery and purity observations. These outputs are influenced by the nonlinear vapour–liquid separation behaviour in the flash separator.

Short Neural Network Validation Parity Plots

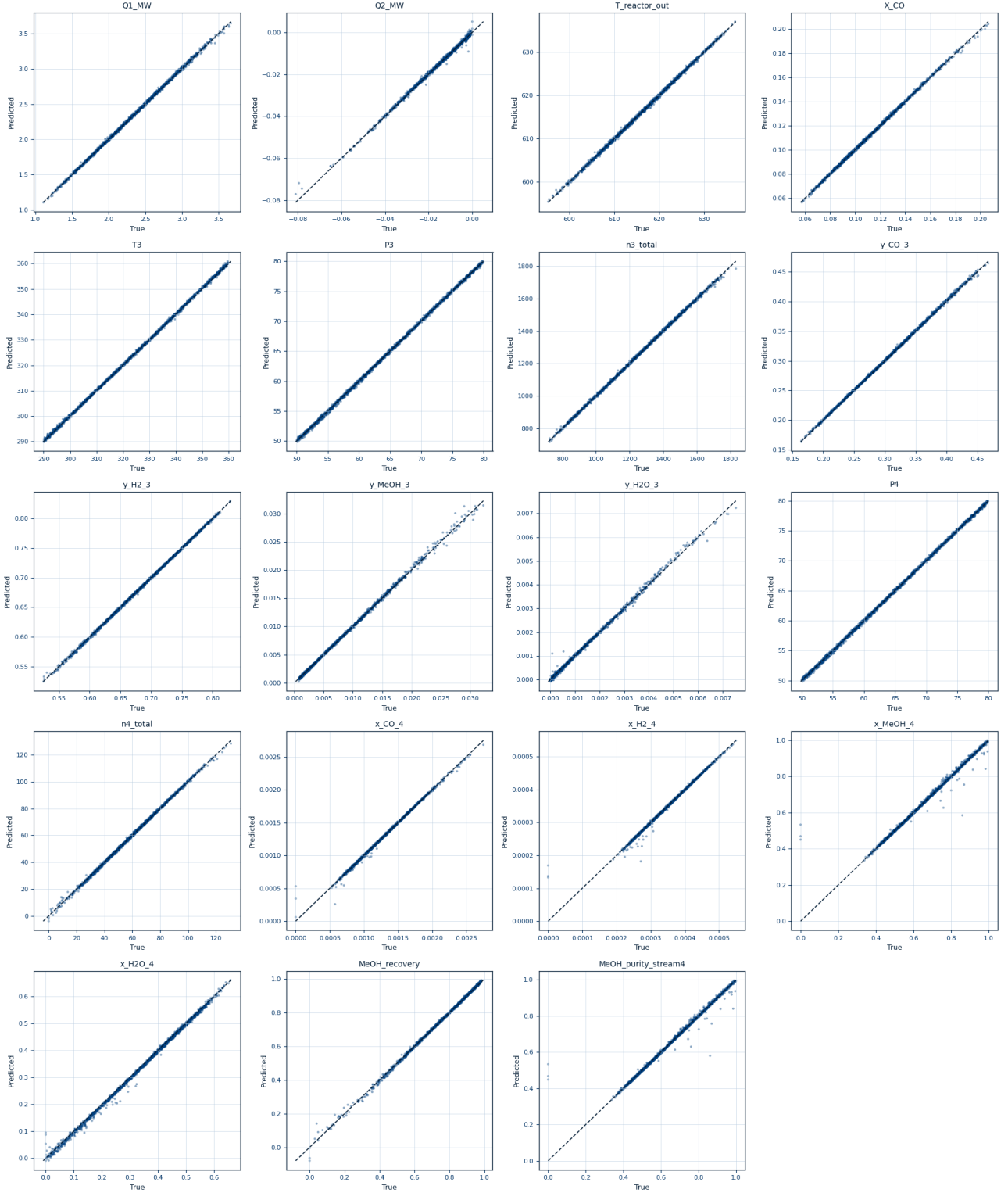


Figure 23: Parity plots for the Short Neural Network..

3.3.2 Long Neural Network

The Long Neural Network was trained using two hidden layers with 124 neurons per layer. The training history is shown in Figure 24: the training and validation losses decrease rapidly during the early training epochs and remain low during the remaining optimization process. The validation curve remains close to the training curve, indicating stable learning behaviour.

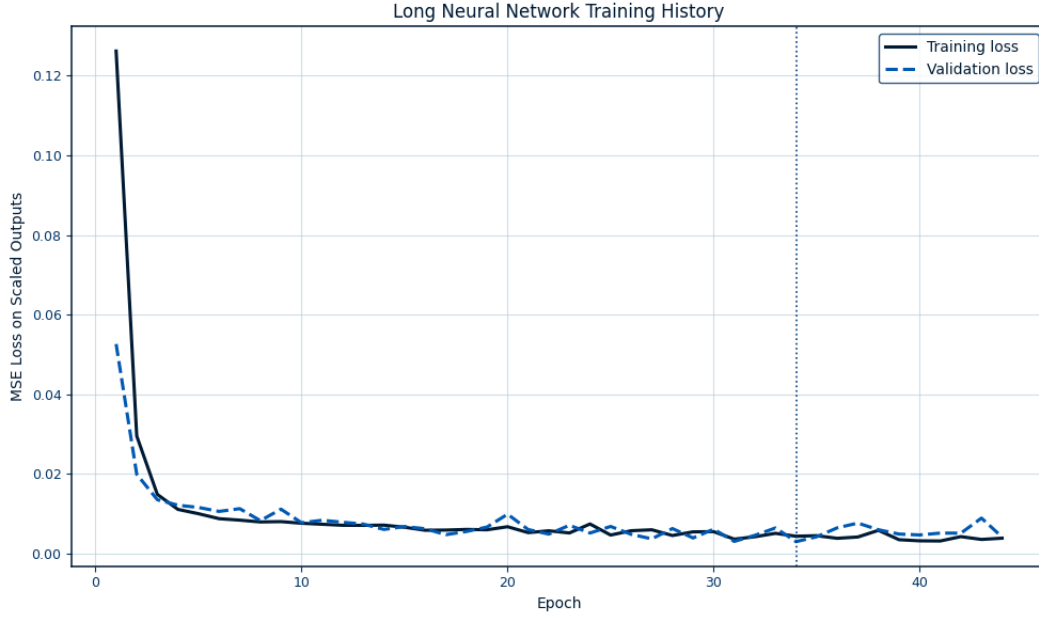


Figure 24: Training and validation loss histories for the Long Neural Network.

The average performance metrics of the Long Neural Network are reported in Table 7. The model achieved a validation mean R^2 of 0.99652 and a median validation R^2 of 0.99825. The minimum validation R^2 of 0.98577 also indicates that the model reproduces all outputs with consistently high accuracy.

Table 7: Average performance metrics of the Long Neural Network.

Dataset	Mean MAE	Mean RMSE	Mean R^2	Median R^2	Minimum R^2
Training	0.391424	0.510420	0.99652	0.99825	0.98577
Validation	0.402370	0.529171	0.99581	0.99787	0.98164

The per-output metrics are shown in Figure 25. The model accurately predicts the process temperatures, pressures, gas-phase compositions, and total stream flow rates. The largest absolute RMSE and MAE values occur for $n_{3,\text{total}}$ and $n_{4,\text{total}}$, which is expected because these outputs have wider numerical ranges than composition variables.

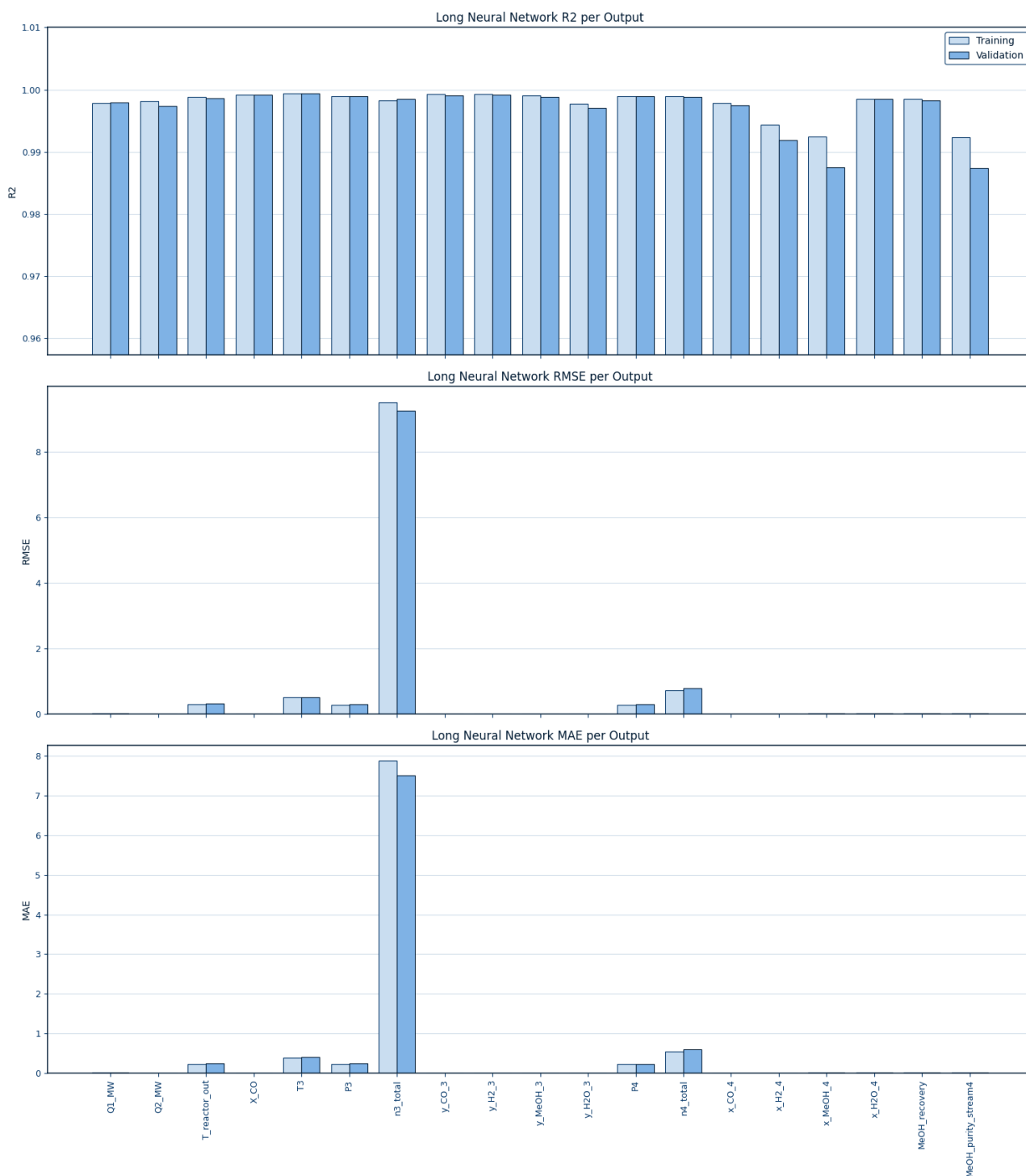


Figure 25: R^2 , RMSE, and MAE values for the Long Neural Network per output.

The validation parity plots are reported in Figure 26. The predicted values are generally aligned with the ideal diagonal line over the complete range of the main process variables. Minor deviations remain for low liquid-phase mole fractions and for the lower range of methanol-related outputs. These regions correspond to conditions where small variations in the flash-separator state can produce relatively large changes in liquid-phase composition.

Long Neural Network Validation Parity Plots

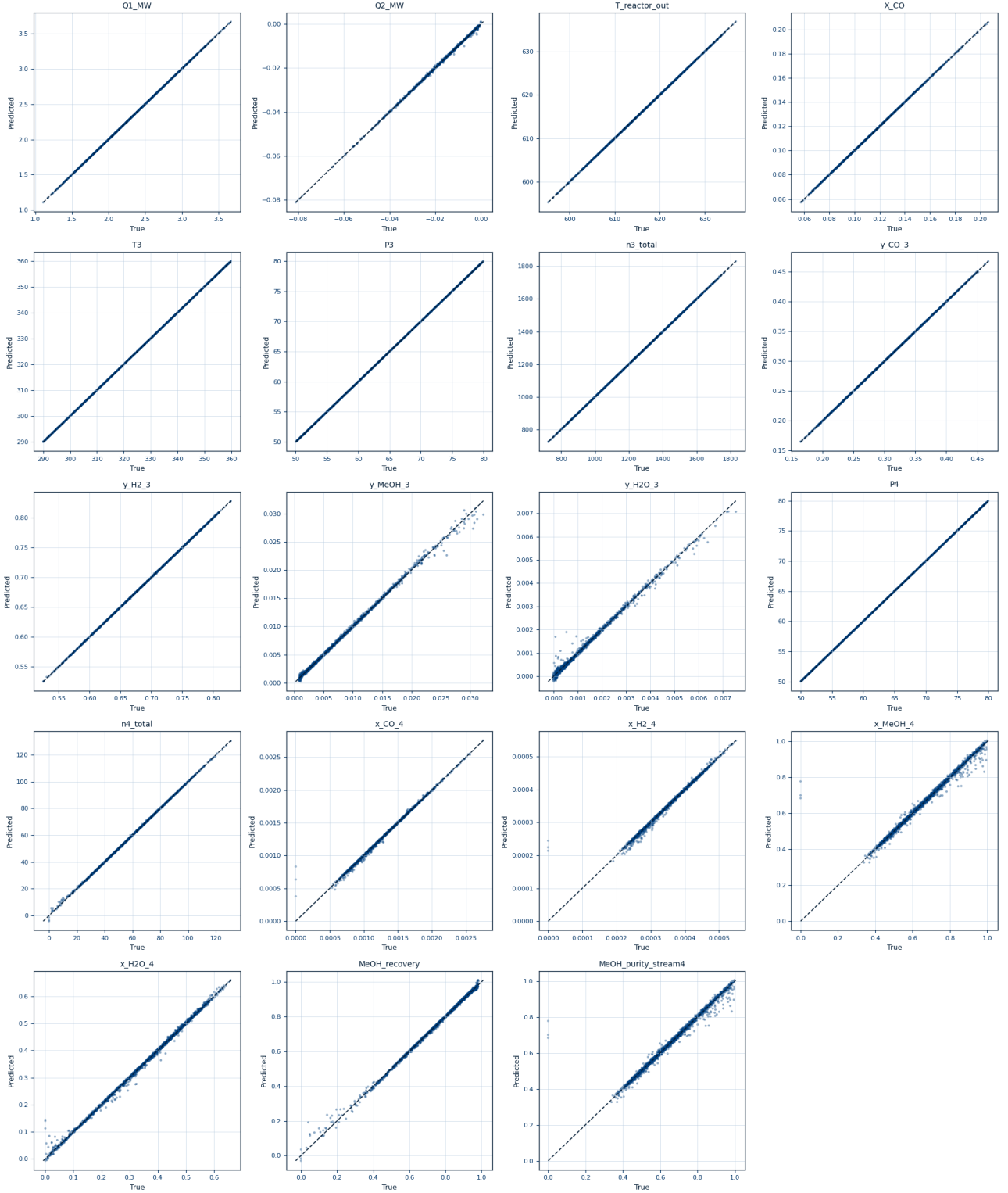


Figure 26: Parity plots for the Long Neural Network.

3.3.3 Hybrid Polynomial Ridge–Neural Network Model

The hybrid model combines the Polynomial Ridge surrogate with a residual neural-network correction. The Polynomial Ridge model provides the baseline prediction of the process response, while the neural network estimates the residual difference between the baseline prediction and the reference output. The final prediction is obtained by adding the neural-network residual correction to the Polynomial

Ridge prediction.

The residual neural-network training history is shown in Figure 27. Both losses decrease during training, while the validation loss gradually approaches a plateau. The final model weights were selected at the minimum validation loss, as indicated by the vertical line in the figure.

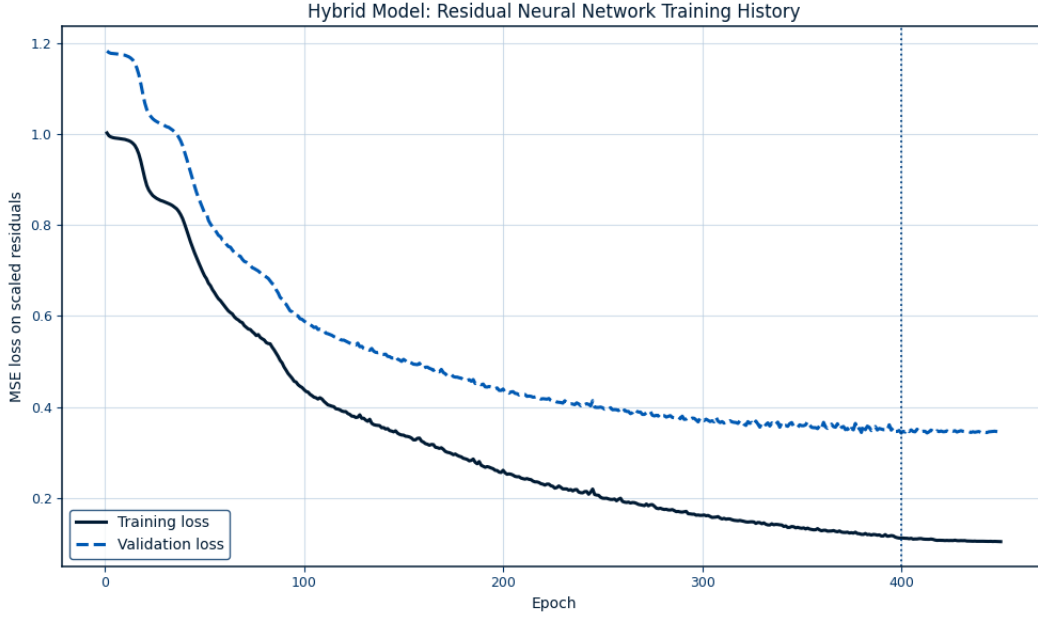


Figure 27: Training and validation loss histories for the Hybrid Neural Network model.

The average performance metrics are reported in Table 8. The hybrid model obtained a validation mean MAE of 0.008594 and a validation mean RMSE of 0.023834. On the final test dataset, the model achieved a mean MAE of 0.007993, a mean RMSE of 0.022971, and a mean R^2 of 0.99307.

Table 8: Average performance metrics of the Hybrid Neural Network model.

Dataset	Mean MAE	Mean RMSE	Mean R^2	Median R^2	Minimum R^2
Validation	0.008594	0.023834	0.99643	0.99990	0.97582
Test	0.007993	0.022971	0.99307	0.99994	0.95516

Figure 28 summarizes the individual R^2 , RMSE, and MAE values for the hybrid Polynomial Ridge–Neural Network model on the validation and test datasets. The model provides R^2 values close to unity for the main process variables, including the heat duties, reactor outlet temperature, CO conversion, stream temperatures and pressures, total stream flow rates, and the principal vapour-phase compositions. The validation and test bars are generally very close, indicating stable predictive behaviour on previously unseen data.

The largest absolute RMSE and MAE values are associated with $n_{3,\text{total}}$ and $n_{4,\text{total}}$. This is expected because these total molar flow rates have substantially larger numerical ranges than mole fractions, conversion, recovery, and purity variables. Their absolute errors should therefore be interpreted together with the corresponding R^2 values, which remain very close to unity.

The liquid-phase outputs show a wider spread in their test-set R^2 values than the main reactor and vapour-stream variables. In particular, $x_{\text{CO},4}$, $x_{\text{H}_2,4}$, $x_{\text{MeOH},4}$, and MeOH purity are relatively more sensitive to local nonlinearities. Nevertheless, their predictive performance remains high, with the

lowest test R^2 values close to 0.95. This behaviour is consistent with the phase-equilibrium effects associated with the flash separation stage, where small changes in operating conditions can lead to comparatively large relative changes in liquid-phase composition and product purity.

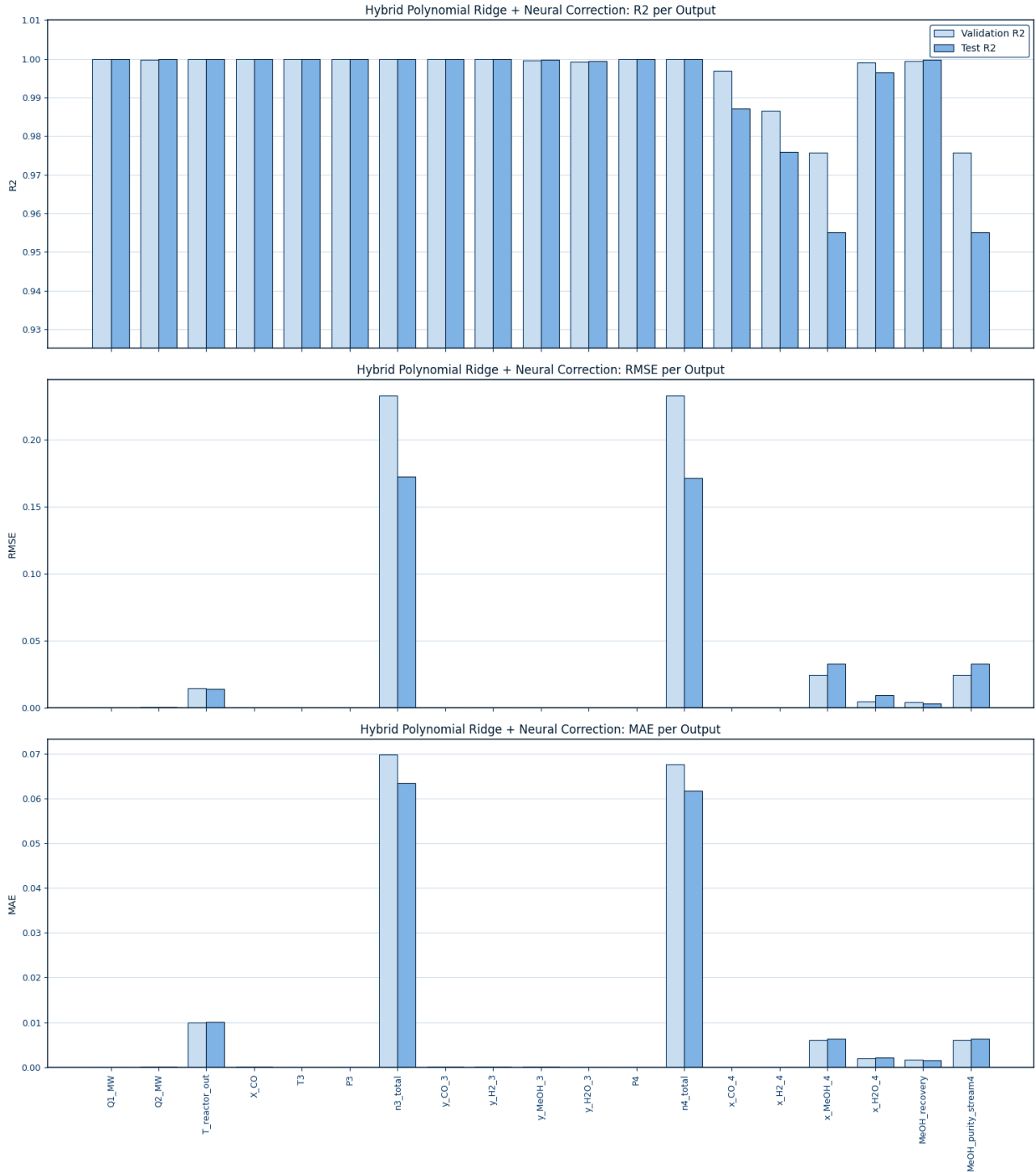


Figure 28: R^2 , RMSE, and MAE values for the Hybrid Neural Network model per output.

The validation parity plots for the hybrid model are shown in Figure 29. The predicted values closely follow the ideal diagonal line for the majority of the output variables, including the reactor outputs, pressure variables, total stream flow rates, and the main vapour-phase compositions. A limited number of deviations remain at very low liquid-phase mole fractions and at the lower range of methanol purity, where the flash-separator behaviour is more sensitive to small changes in the operating conditions.

Hybrid Model Test Parity Plots for Outputs

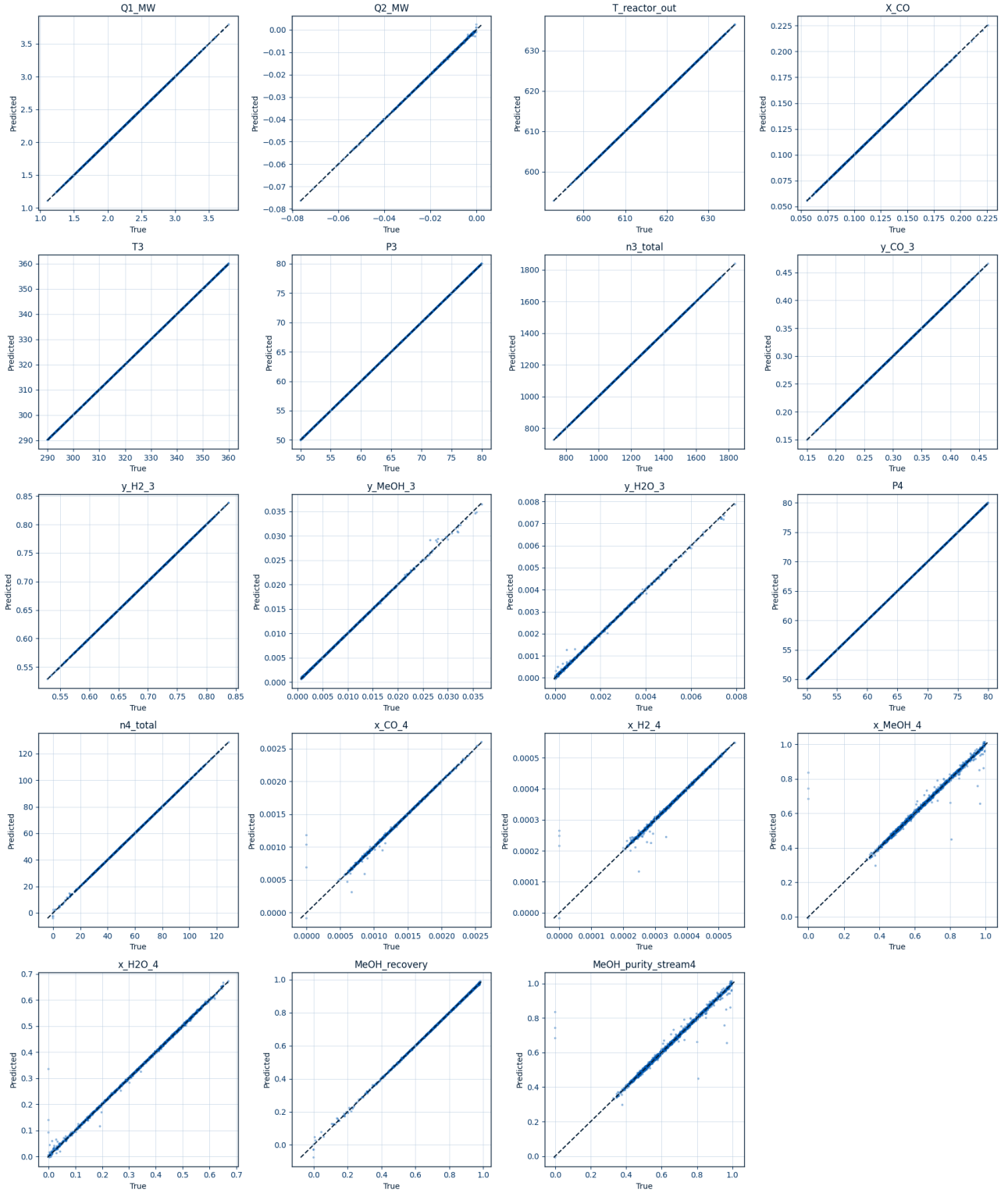


Figure 29: Parity plots for the Hybrid Neural Network model per output

3.4. Comparison between architectures

The three surrogate-model architectures were evaluated in order to identify the most reliable model for the subsequent constrained optimization stage. To recall the investigated cases were the following:

- **Short Neural Network**

- Long Neural Network
- Hybrid Polynomial Ridge + Residual Neural Network

All models were evaluated using the same train-validation split, obtained from the original training dataset with a fixed random seed. The independent test set was retained exclusively for the final assessment of generalization performance, this guarantee a fair comparison since the test set was not used during model selection. The comparison considered both global performance across the nineteen process outputs and performance on the outputs directly relevant to the optimization problem (Q_1 , Q_2 , MeOH recovery, MeOH purity₄) which were given particular importance because they will determine the optimization objective and constraints in the next section.

The performance indicators are reported in table 9, where MAE and RMSE quantify the absolute magnitude of the prediction error, with RMSE penalizing large deviations more strongly. Since the nineteen outputs have different physical units and numerical ranges, the NRMSE was also calculated by normalizing the RMSE of each output by its corresponding range, allowing prediction errors associated with different outputs to be compared on a common basis.

Dataset	Model	MAE	RMSE	NRMSE	Mean R^2	Opt. MAE	Opt. RMSE	Opt. NRMSE	Opt. R^2
Validation	Short NN	0.364581	0.457140	0.008884	0.995520	0.005563	0.012091	0.011689	0.992190
Validation	Long NN	0.376447	0.504651	0.009901	0.995449	0.006095	0.012735	0.012135	0.992662
Validation	Hybrid Poly+Res	0.008594	0.028384	0.005067	0.996431	0.001925	0.007182	0.007691	0.993776
Test	Short NN	0.349969	0.443246	0.009417	0.994544	0.005338	0.012247	0.011882	0.991096
Test	Long NN	0.383716	0.502630	0.011076	0.993090	0.006107	0.014061	0.013423	0.989298
Test	Hybrid Poly+Res	0.007993	0.022971	0.006792	0.993072	0.001990	0.009006	0.009465	0.988690

Table 9: Comparison of the predictive performance of the NN architectures

The global validation results show that the Hybrid Polynomial Ridge + Residual NN achieved the lowest average errors among the three architectures. Its validation MAE and RMSE were equal to (0.008594) and (0.028384), respectively, substantially lower than the corresponding values obtained by the Short NN ((0.364581) and (0.457140)) and the Long NN ((0.376447) and (0.504651)). The hybrid model also achieved the lowest global validation NRMSE, equal to (0.005067), compared with (0.008884) for the Short NN and (0.009901) for the Long NN.

A similar ranking was obtained on the independent test set: the Hybrid model achieved a test MAE of (0.007993), a RMSE of (0.022971), and a NRMSE of (0.006792). These results confirm that the improved performance of the hybrid architecture was not limited to the validation data.

The comparison is even more relevant when considering only the optimization-related outputs: on the validation set, the Hybrid model achieved an optimization-output NRMSE of (0.007691), which was lower than the values obtained by the Short NN ((0.011689)) and Long NN ((0.012135)). Therefore, relative to the Short NN, the hybrid architecture reduced the validation NRMSE associated with the optimization-relevant outputs by approximately (35%) and relative to the Long NN, the reduction was approximately (37%). The test-set results confirmed the same ordering, the Hybrid model obtained an optimization-output NRMSE of (0.009465), while the Short NN and Long NN obtained values of (0.011882) and (0.013423), respectively. Therefore, the hybrid surrogate remained the most accurate model on the independent test data for the quantities that govern the optimization problem. Although the Short NN achieved slightly higher test R^2 values than the Hybrid model for some aggregated metrics, this did not change the model-selection decision, since this coefficient is a measure of the

fraction output variance explained by the model is less relevant in our case than NRMSE.

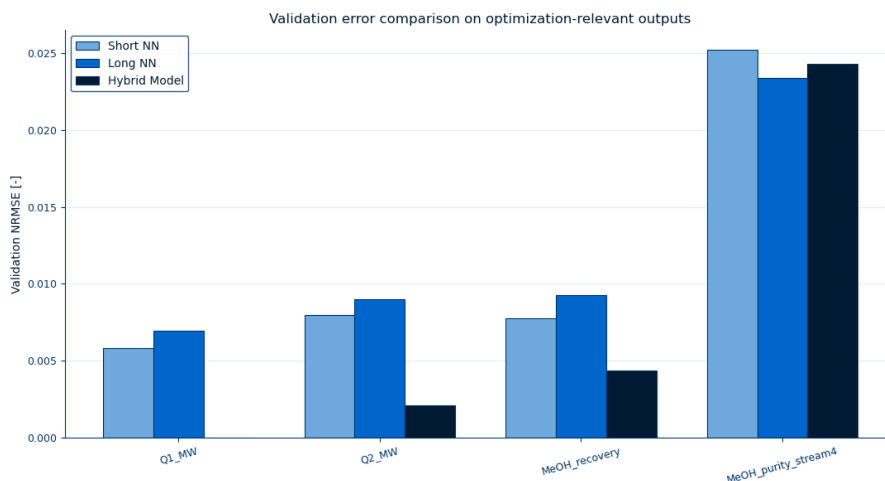


Figure 30: Validation error comparison on optimization of the most relevant outputs

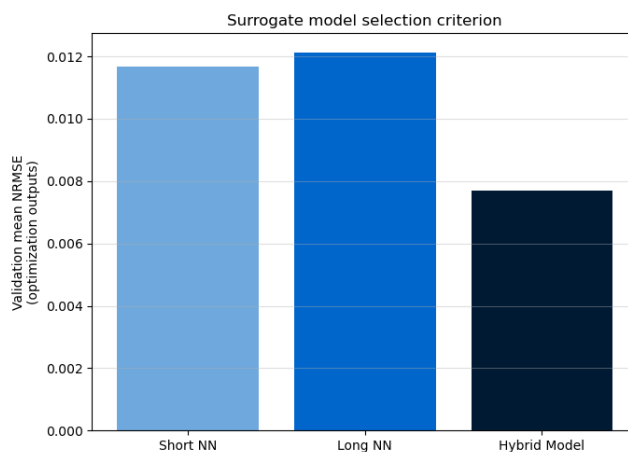


Figure 31: Surrogate model selection criterion

A good point to underline is that the Long NN did not improve the prediction performance with respect to the Short NN, this suggest that simply increasing the network depth and width doesn't guarantee an improvement under the adopted training conditions, having a higher number of trainable parameters (as the Long does with respect to the Short) may require additional hyperparameter tuning, such as a modified learning rate, longer training, weight decay, dropout, or a different batch size, in order to exploit its additional representational capacity.

The superior performance of the Hybrid model can be explained by its decomposition of the learning task: the Polynomial Ridge component captures the dominant smooth nonlinear behaviour of the reaction-separation process, and the residual neural network is then required to learn only the discrepancy between the Polynomial Ridge prediction and the reference data. This structure reduces the complexity of the function that must be learned by the neural network since instead of reconstructing the complete process response from the input variables, the residual neural network only focuses on the nonlinear effects that are not adequately represented by the polynomial baseline.

4. Application

The final stage of the project consisted of using the trained surrogate model to perform a constrained process optimization to maximize the methanol recovery in stream 4, while satisfying two constraints: the first on the energy duty with overall $|Q_1| + |Q_2| \leq Q_{\max}$ with $Q_{\max} = 1.7$ MW; and the second on the methanol purity of stream 4, imposed by us at 95% to have a value coherent with the simplified scheme of the process.

The best neural network architecture selected to be optimized was the hybrid polynomial ridge with residual neural network model. As said above the choice of the surrogate architecture for optimization was not based on the global performance over all nineteen outputs alone, instead, the selection focused on the outputs directly involved in the optimization problem therefore the one with the lowest NRMSE on optimization-relevant outputs only is the one we decide to further investigate.

The optimization problem was formulated over the nine original process input variables:

$$x = [n_{1,\text{total}}, y_{CO,1}, y_{H_2O,1}, n_{2,\text{total}}, T_1, T_{\text{reactor,in}}, P_R, T_{\text{flash}}, T_4]$$

The heat duties (Q_1) and (Q_2), the methanol recovery, and the methanol purity were not treated as independent decision variables, instead, they were predicted by the surrogate model from the selected input conditions. This is important because the optimizer modifies only the process inputs, while the surrogate model estimates the corresponding process performance.

The optimization was subject to three constraints: the first constraint limited the total heat duty required by the two heat exchangers: $|Q_1| + |Q_2| \leq Q_{\max}$ with $Q_{\max} = 1.7$ MW. The absolute values were used because Q_1 and Q_2 have different signs according to the heating/cooling convention. The second constraint imposed a minimum methanol purity in stream 4: $MeOH_{\text{purity}} \geq 0.95$. This constraint prevents the optimizer from maximizing recovery at the expense of producing a liquid stream with insufficient methanol concentration. The third constraint ensured the physical consistency of the cooling step after the flash: $T_4 \leq T_{\text{flash}} - 2$: this reflects the fact that the second heat exchanger acts as a cooler, therefore, the outlet liquid temperature (T_4) must be lower than the flash temperature and the additional 2 K margin avoids solutions where the cooling duty is physically negligible or inconsistent. All constraints were implemented in the form required by the SLSQP algorithm, namely as inequality functions that must be greater than or equal to zero, at the same time, these quantities were also used after each optimization run to compute the constraint slack (positive indicates that the constraint is satisfied with margin, zero indicates that the constraint is active and a negative slack would indicate violation).

The optimization was implemented through a general prediction function, which received a candidate vector of nine process inputs, reshaped it into the required format and passed it to the surrogate model. Since we're using hybrid model the Polynomial Ridge prediction was first computed, then the residual correction was predicted by the residual neural network, and finally the two contributions were added. For each candidate input point, the surrogate predicted the full vector of nineteen outputs and from this output vector, the code extracted the quantities needed for the optimization:

$$[Q_1, Q_2, |Q_1| + |Q_2|, \text{MeOH recovery}, \text{MeOH purity}_4]$$

These quantities were used to evaluate both the objective function and the constraints allowing the optimizer to explore the input space without directly calling the original simulator. The optimization itself was performed using the SLSQP algorithm, this was chosen because it can handle nonlinear objective functions, nonlinear constraints, and bound constraints on the decision variables. Each of the nine input variables was constrained within its admissible range provided.

Because the surrogate response is nonlinear and the constrained optimization problem may contain local optima, a multi-start strategy was used: first, the code searched the original training dataset for points that already satisfied the constraints, and, among these feasible points, the ones with the highest methanol recovery were used as initial guesses for the optimizer. In addition to it random initial guesses were generated inside the admissible input bounds with the relation constrain between T_4 and T_{flash} imposed. This strategy increased the robustness of the search and reduced the risk of having a final result depending only on the initial starting points.

Each optimization run produced a candidate optimum, for every candidate solution, the surrogate-predicted outputs were used to recompute the energy duty, methanol purity, methanol recovery, and constraint slacks. A candidate was classified as feasible only if all three constraints were satisfied simultaneously and finally among all the feasible candidates the one with the highest predicted methanol recovery was selected as the optimum.

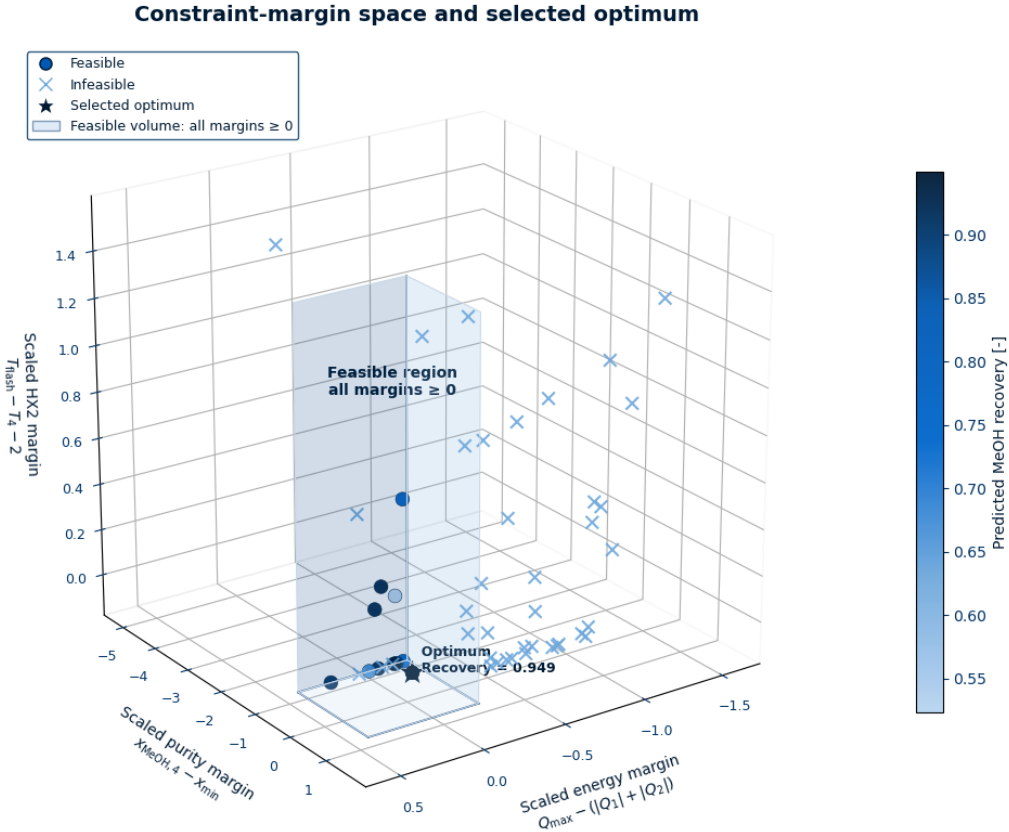


Figure 32: 3D constrain margin space

Figure 32 provides a three-dimensional representation of the candidate solutions generated by the multi-start constrained optimization procedure. The selected operating point (highlighted by the star marker) corresponds to the feasible candidate with the highest predicted methanol recovery among all

optimization runs. The figure shows that the selected optimum lies within the feasible region and close to the HX2 cooling boundary. This confirms that the cooling constraint is active, or nearly active, at the optimum: the optimizer tends to increase T_4 until the maximum temperature compatible with a physically consistent cooling duty is reached. In contrast, the energy-duty and purity constraints retain a positive margin at the selected point. Therefore, the final solution is primarily limited by the thermal relationship between the flash temperature and the HX2 outlet temperature, rather than by the available energy duty or the minimum product-purity requirement.

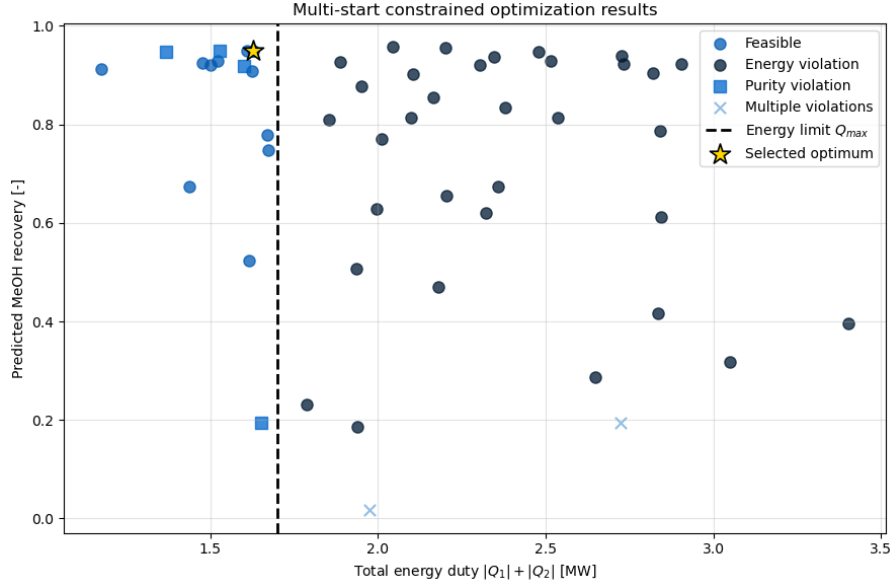


Figure 33: Multi-start constrained optimization results

The multi-start optimization scatter plot showed the recovery achieved by the different optimization runs as a function of the total energy duty, distinguishing feasible and infeasible solutions. This plot made it possible to see that some points with high recovery were not acceptable because they violated one or more constraints. The visualization of recovery versus energy duty is particularly useful because it shows the trade-off between performance and energy consumption, the selected optimum lies below the energy limit and close to the highest recovery values, which supports the validity of the final decision.

Therefore the optimal operating point for our system is the following:

Input	Optimal value
n1_total	1130.134590
y_CO_1	0.429373
y_H2O_1	0.000809
n2_total	95.344718
T1	316.634563
T_reactor_in	478.259853
P_R	56.111722
T_flash	292.010054
T4	290.010054

Table 10: Optimal operating conditions identified by the surrogate-based constrained optimization.

The output for the optimal operating point, evaluated through the surrogate model are the following:

Output	Predicted value
Q1_MW	1.626291
Q2_MW	-0.001165
T_reactor_out	604.943061
X_CO	0.095654
n3_total	1087.493010
n4_total	45.131130
x_MeOH_4	0.973412
x_H2O_4	0.021858
MeOH_recovery	0.949311
MeOH_purity_stream4	0.973446

Table 11: Relevant surrogate-predicted process outputs at the optimal operating point.

A nearest-neighbour analysis was performed in the standardized input space to assess whether the optimized operating point was located within a well-sampled region of the training domain. The closest training sample had a scaled Euclidean distance of $1.26 * 10^{-7}$, indicating that the selected optimum practically coincides with an existing dataset point, therefore, the optimization result does not rely on extrapolation of the surrogate model. Moreover, the surrogate predictions were consistent with the corresponding dataset outputs, particularly for the total heat duty and methanol recovery. This supports the reliability of the selected operating point within the explored design space.

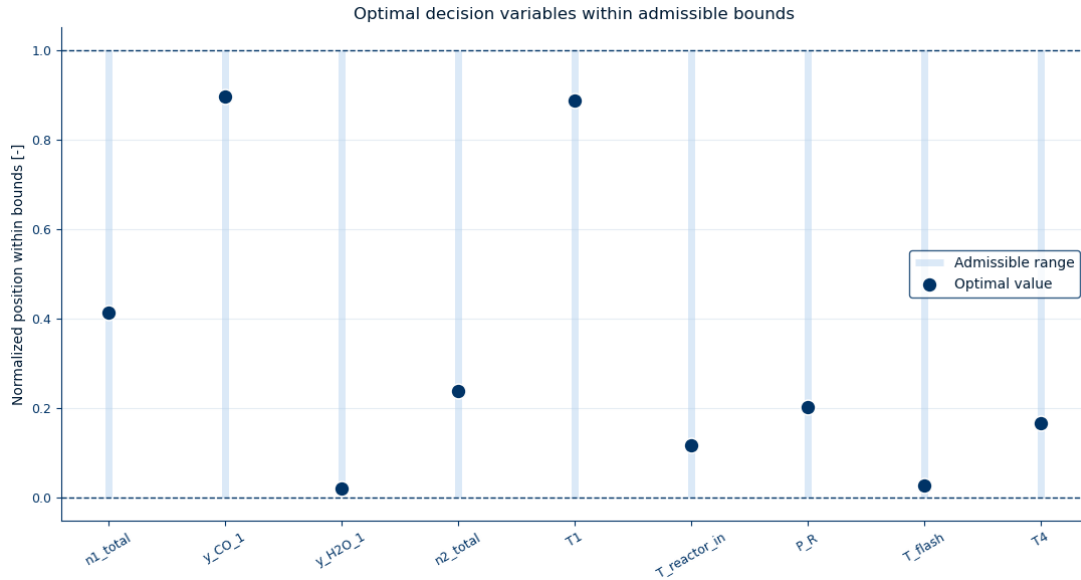


Figure 34: Optimal decision variables within admissible bounds

The normalized decision-variable plot showed where the optimal operating point lies within the admissible bounds of each input. If some variables are located close to 0 or 1 it means that the optimizer is pushing them toward the edge of the studied domain. This can be interpreted in two ways: on one hand, it may indicate a clear physical direction for improving the process; on the other hand, it may also suggest that the optimum could lie outside the original design space if the bounds were extended. For this reason, variables close to their bounds should be treated carefully and discussed as potential candidates for further investigation. To have not only graphical representation but also numerical results the constraint satisfaction table summarized the value, limit, slack, and status of each constraint at the selected optimum.

Constraint	Condition	Value	Limit	Slack	Status
Energy duty	$ Q_1 + Q_2 \leq Q_{\max}$	1.6275 MW	≤ 1.7000 MW	0.0725 MW	Satisfied
Methanol purity	$x_{\text{MeOH},4} \geq x_{\min}$	0.9734	≥ 0.9500	0.0234	Satisfied
HX2 cooling	$T_4 \leq T_{\text{flash}} - 2$	290.01 K	≤ 290.01 K	0.0000 K	Active

As we can see from the results of the optimum operating point the energy constraint was satisfied with a positive slack (around 0.0725 MW) meaning that the selected solution used slightly less heat duty than the maximum allowed. The methanol purity constraint was also satisfied with a positive margin, approximately (0.0234), which means that the predicted methanol purity was above the minimum required value of 0.95. Therefore, the selected solution did not maximize recovery by sacrificing product quality, this is an important result because without the purity constrain the optimizer could potentially favor operating conditions that increase the total methanol recovered but dilute the liquid product. The HX2 cooling constraint, instead, was essentially active at the optimum having a slack near zero, this means that the optimizer pushed (T_4) to the maximum value allowed by the cooling condition. Physically, this suggests that the surrogate model predicted a benefit from operating the post-flash liquid outlet temperature as high as possible while still satisfying the requirement that HX2 behaves as a cooler. This is a meaningful result because it indicates that the thermal condition after the flash

is one of the limiting factors of the optimum. This would also suggest that the constrain with the zero slack is in a critical position in case of a local error of the surrogate predictions, therefore in real life plant if the limit is a strict physical limit (example for safety reasons related to the unit) it could be a good improvement to introduce a safety margin in the constrains.

Overall, the optimization results show that the surrogate-based procedure was able to identify a feasible high-recovery operating point while respecting the energy, purity, and physical cooling constraints. The use of a multi-start strategy improved the robustness of the search by reducing the dependence on a single initial guess. The selected optimum is therefore the best feasible candidate found by the SLSQP search using the selected hybrid surrogate.